

Lecture 12: Quantum mechanics and molecular structure-3

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Classroom: Teaching Center 401

Wednesday/Friday: 10:15-11:00; 11:10-11:55

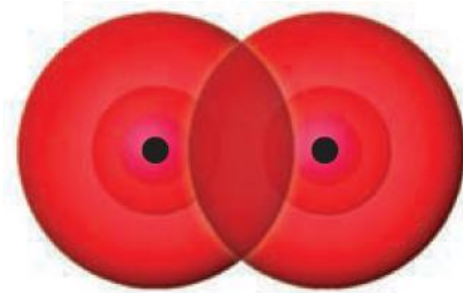
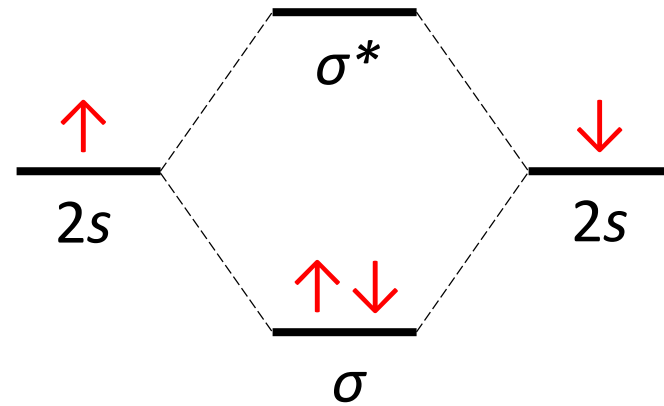
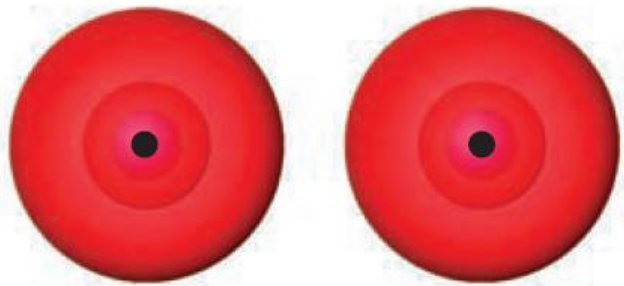
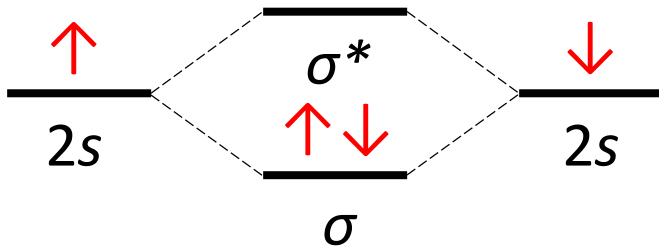
2025/11/21

Major points of last lecture

- **Quantum mechanics and molecular structure**
 5. **Homonuclear Diatomic Molecules: Second-Period Atoms**
 6. **Heteronuclear Diatomic Molecules**
 7. **Summary Comments for the LCAO Method and Diatomic Molecules**
 8. **Heteronuclear Diatomic Molecules**
 9. **Summary Comments for the LCAO Method and Diatomic Molecules**

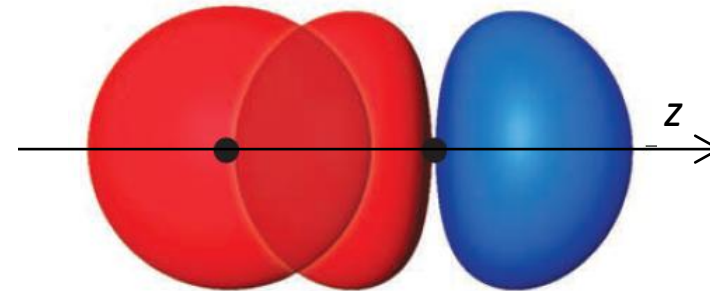
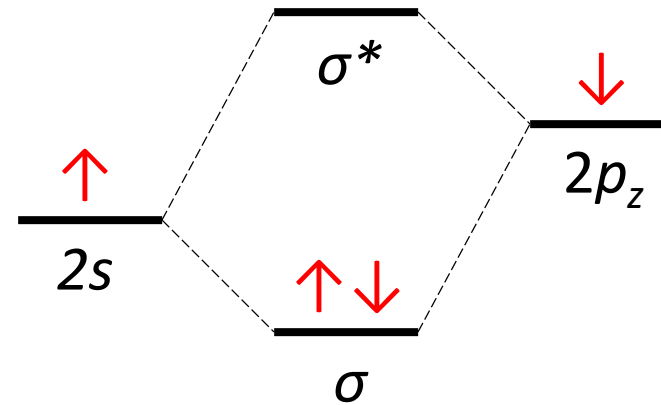
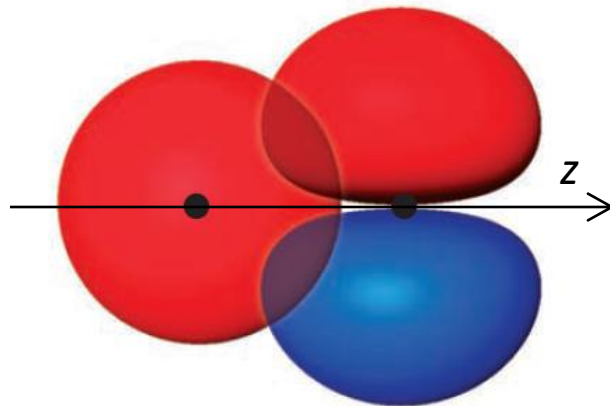
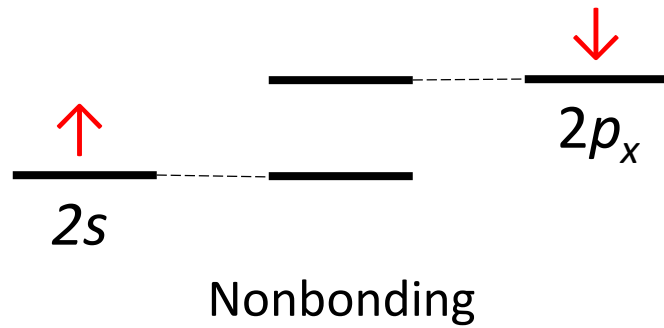
Factors for Effective Bonding

① Maximum orbital overlap



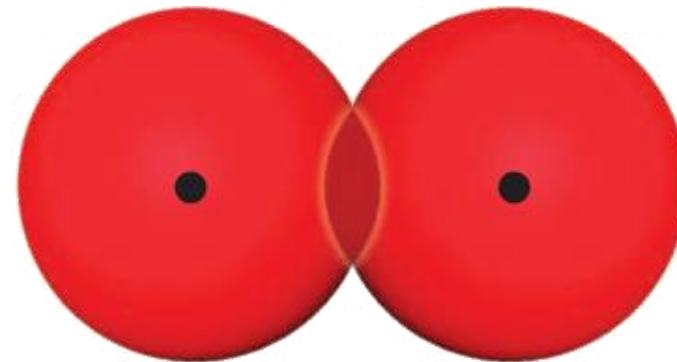
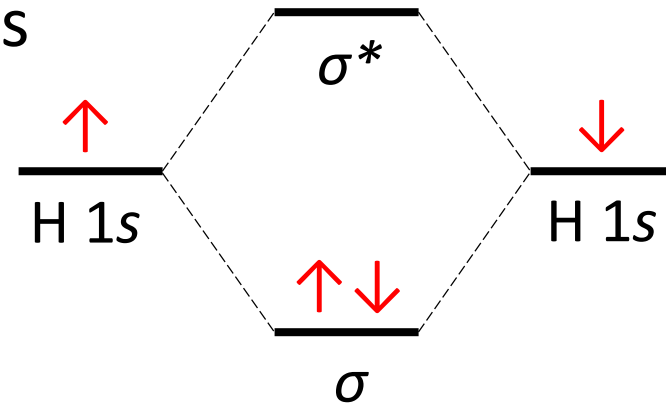
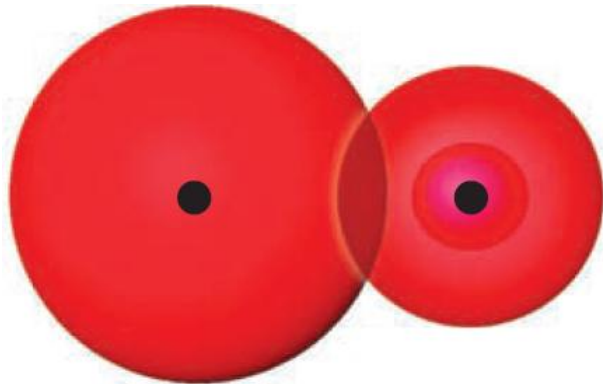
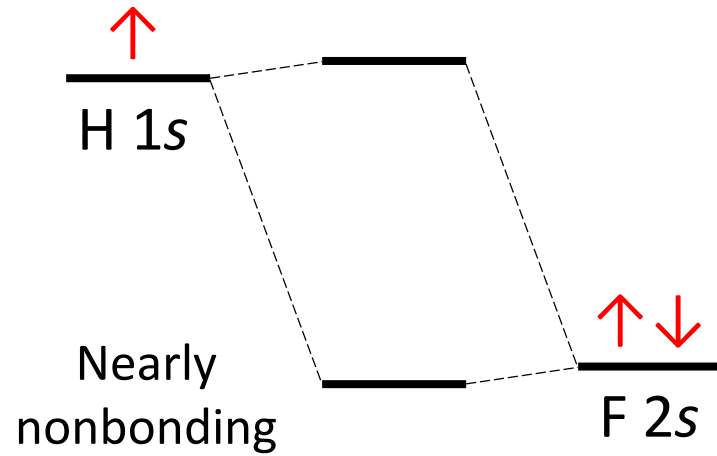
Factors for Effective Bonding

② Compatible orbital symmetries

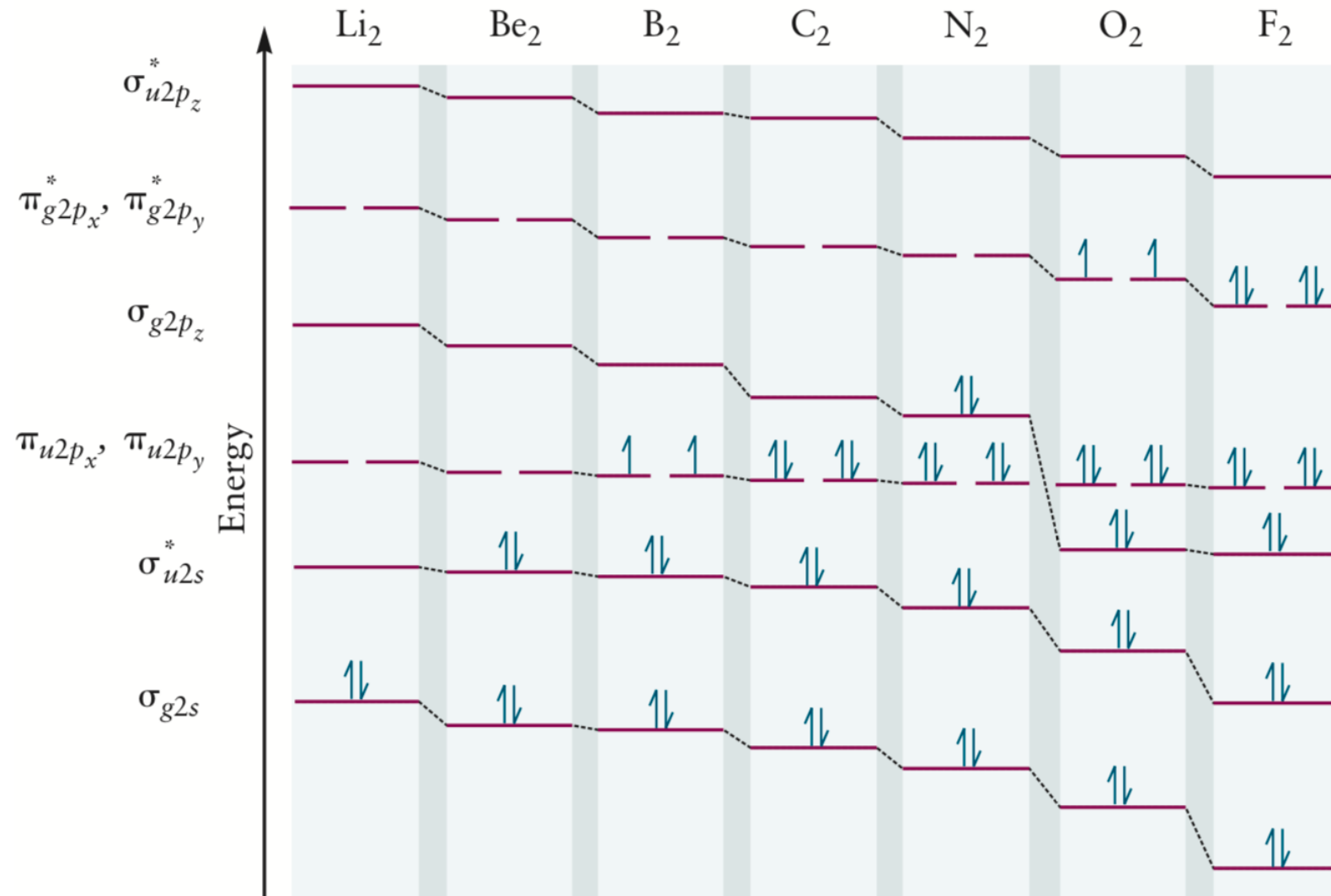


Factors for Effective Bonding

③ Approximate atomic orbital energies



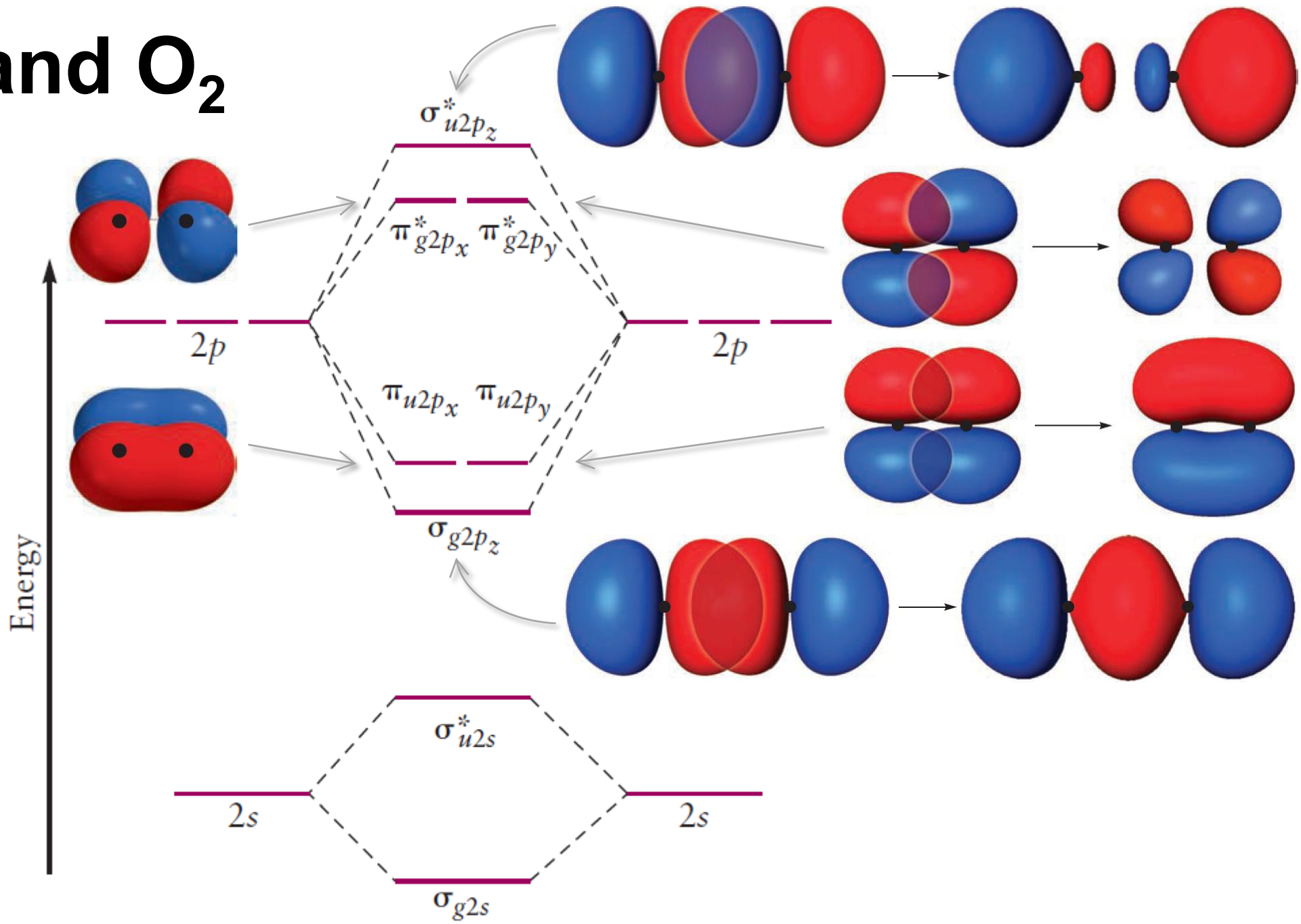
Homonuclear Diatomic Molecules: Second-Period Atoms



increased nuclear charge (Z_{eff}) felt by the electrons in the heavier elements.

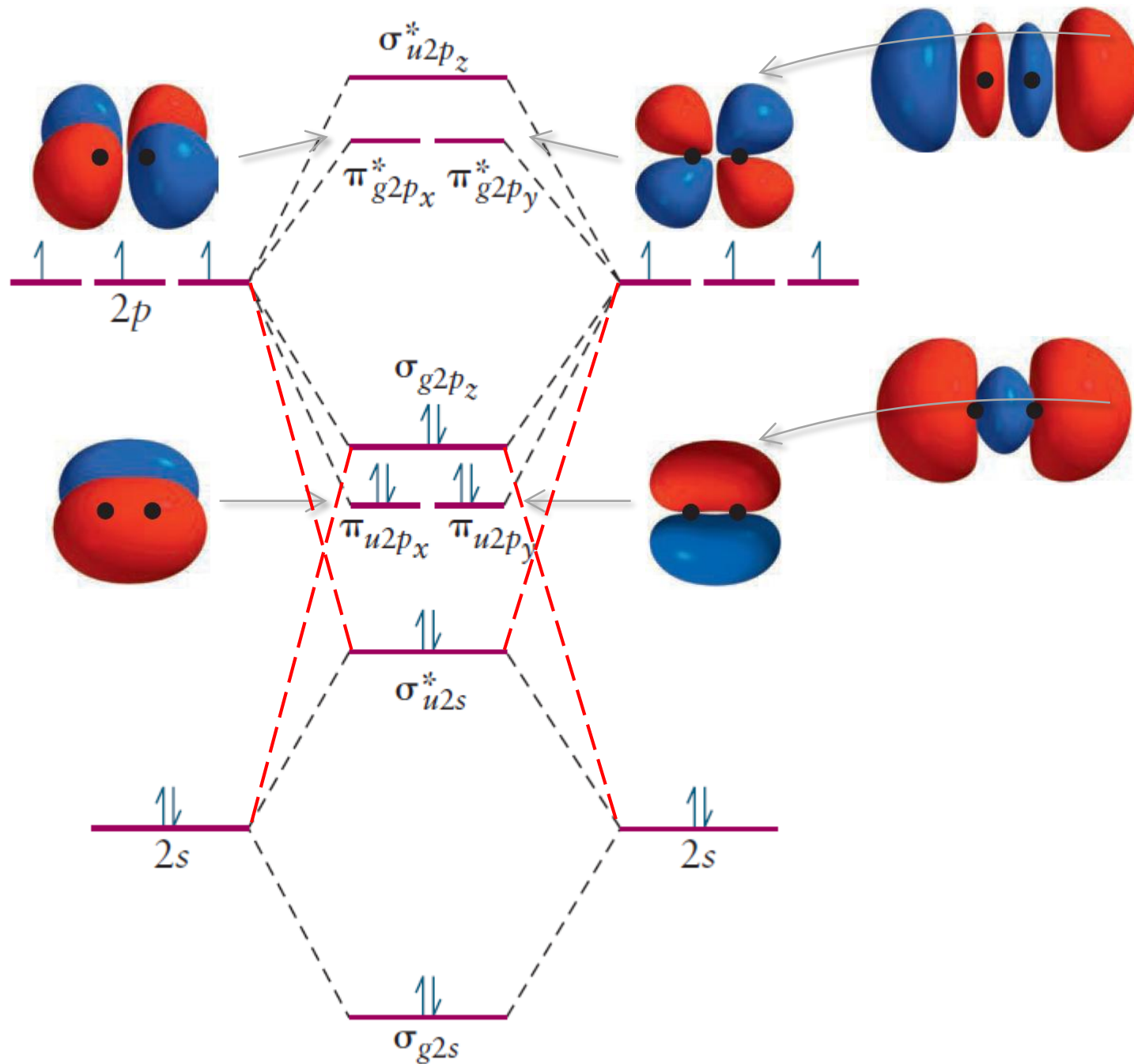
The electrons have been placed in MOs according to the Aufbau principle.

F₂ and O₂

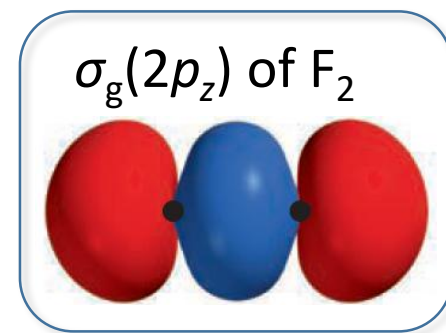


N₂

Energy ↑



VS



Heteronuclear Diatomic Molecules

1. A different set of labels: **heteronuclear diatomic molecules lack the inversion symmetry of homonuclear diatomic molecules.** (drop the g and u subscripts on the MO labels)

2. AOs on the participating atoms now correspond to **different energies: $C_A \neq C_B$**

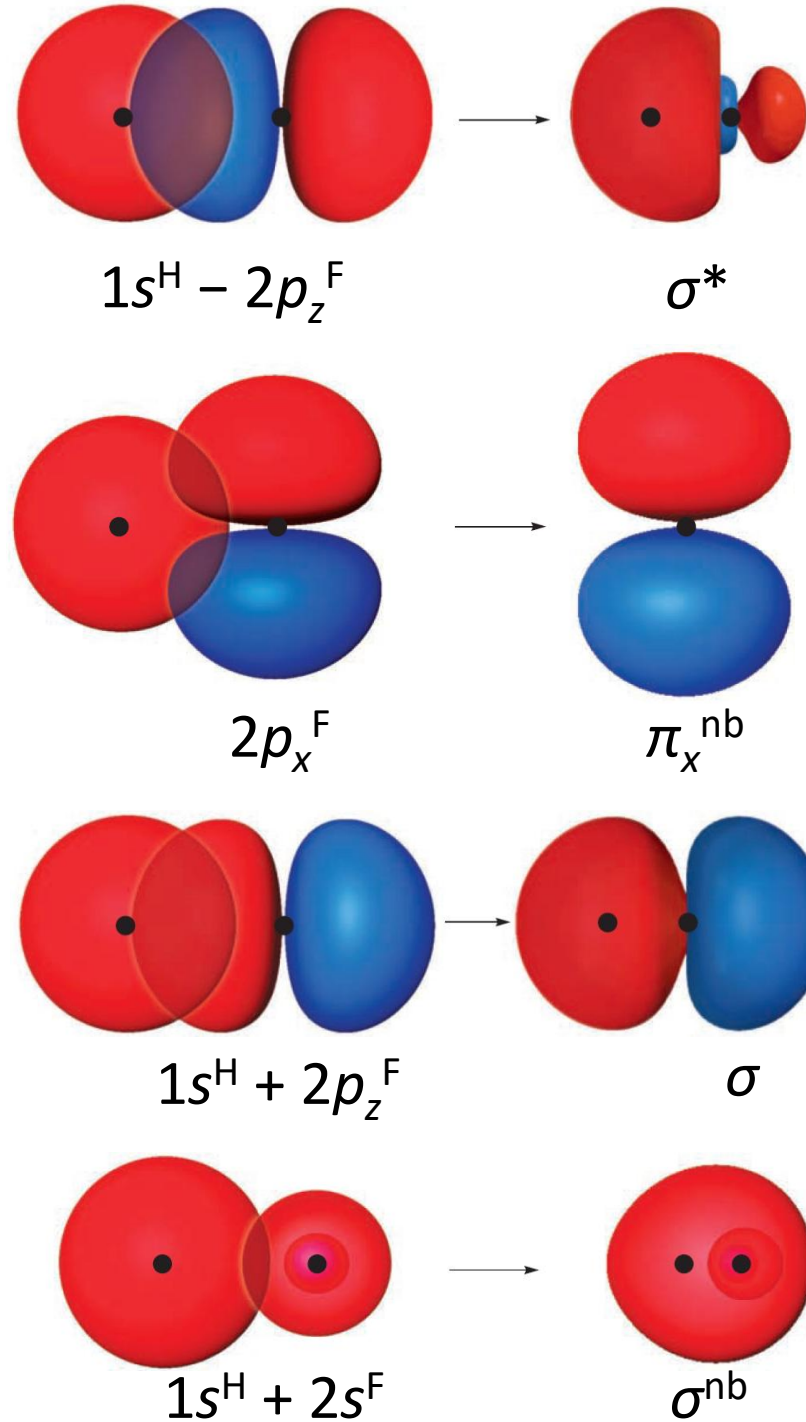
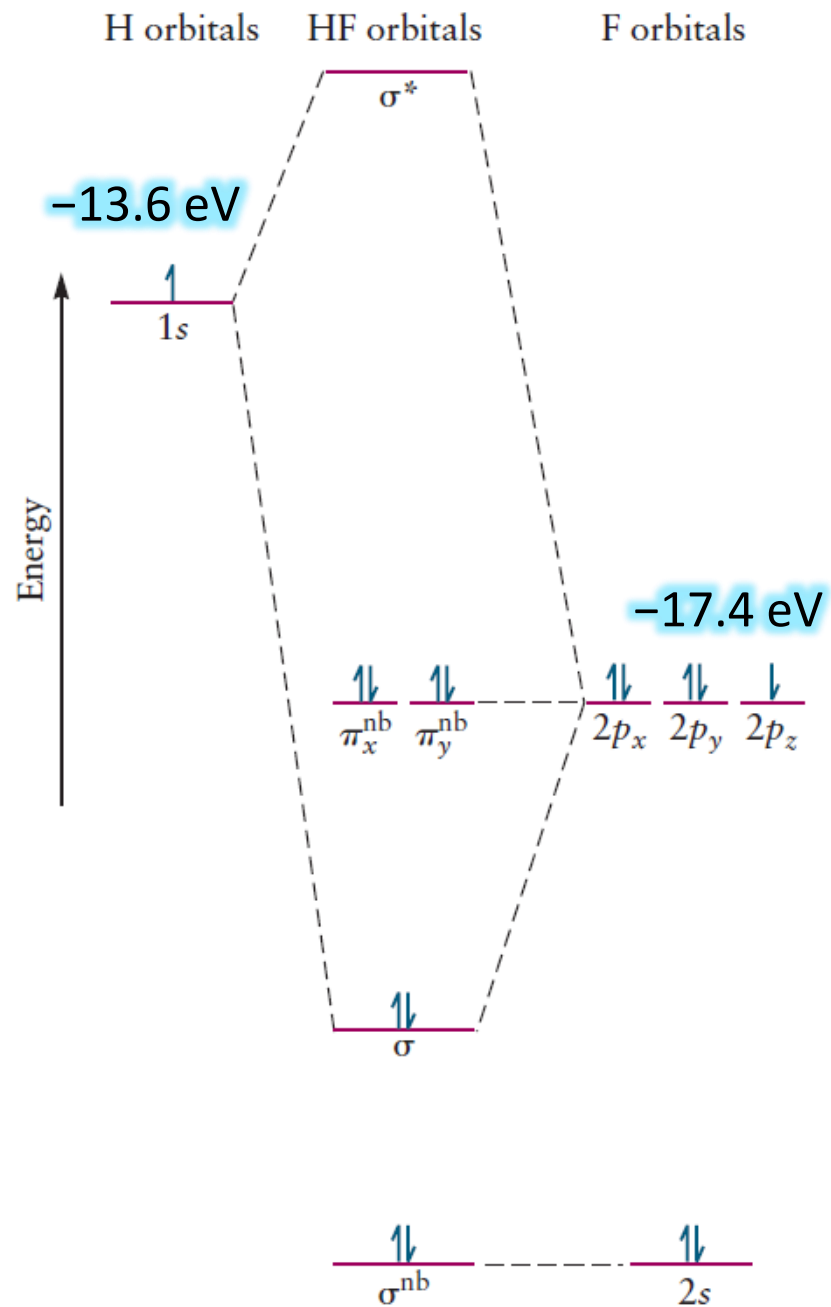
$$\sigma_{2s} = C_A 2s^A + C_B 2s^B$$

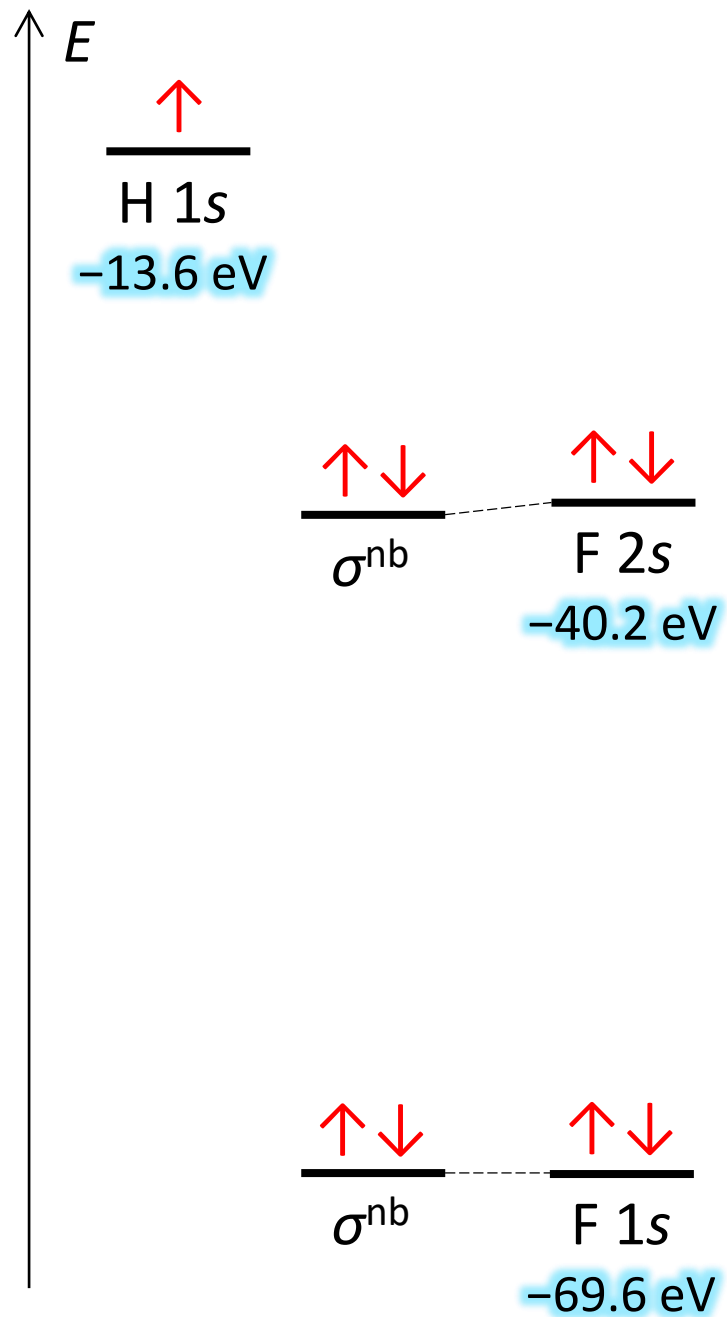
$$\sigma_{2s}^* = C'_A 2s^A - C'_B 2s^B$$

- Atom B is more electronegative than atom A: $C_B > C_A$ for the bonding σ MO
(and the electron spends more time on the electronegative atom)
- $C'_A > C'_B$ for the higher energy σ^* MO and it more closely resembles a $2s^A$ AO

CO: carbon monoxide

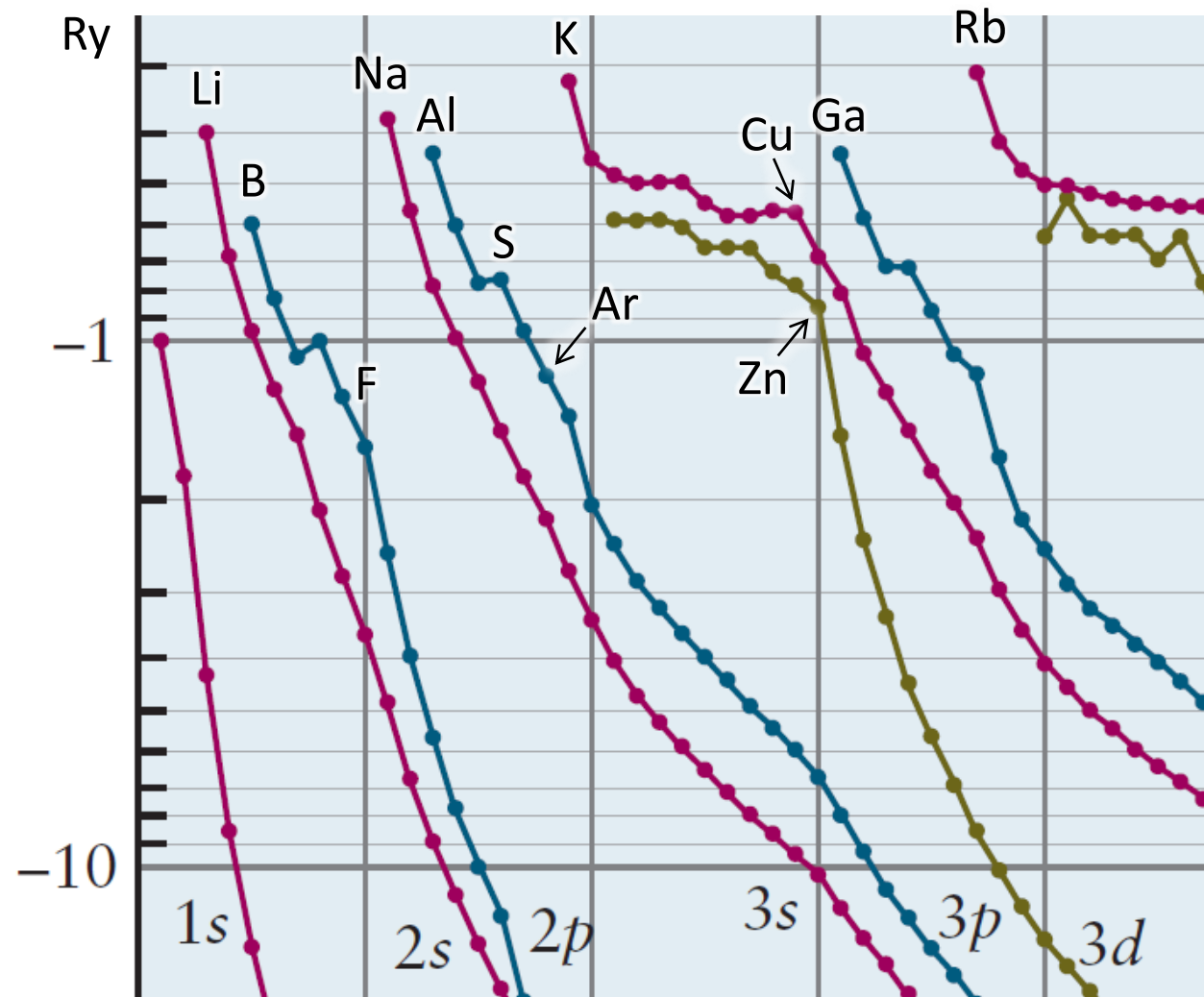
HF





Unit:

Fig. 5.26, OGB8 p.210



LCAO-MO: Summary

Atomic orbitals and molecular orbitals

- Can be linearly combined
- Obey the Aufbau Principle

Rules of bonding

- Maximum orbital overlap
- Compatible orbital symmetries
- Approximate atomic orbital energies

Major points of today's lecture

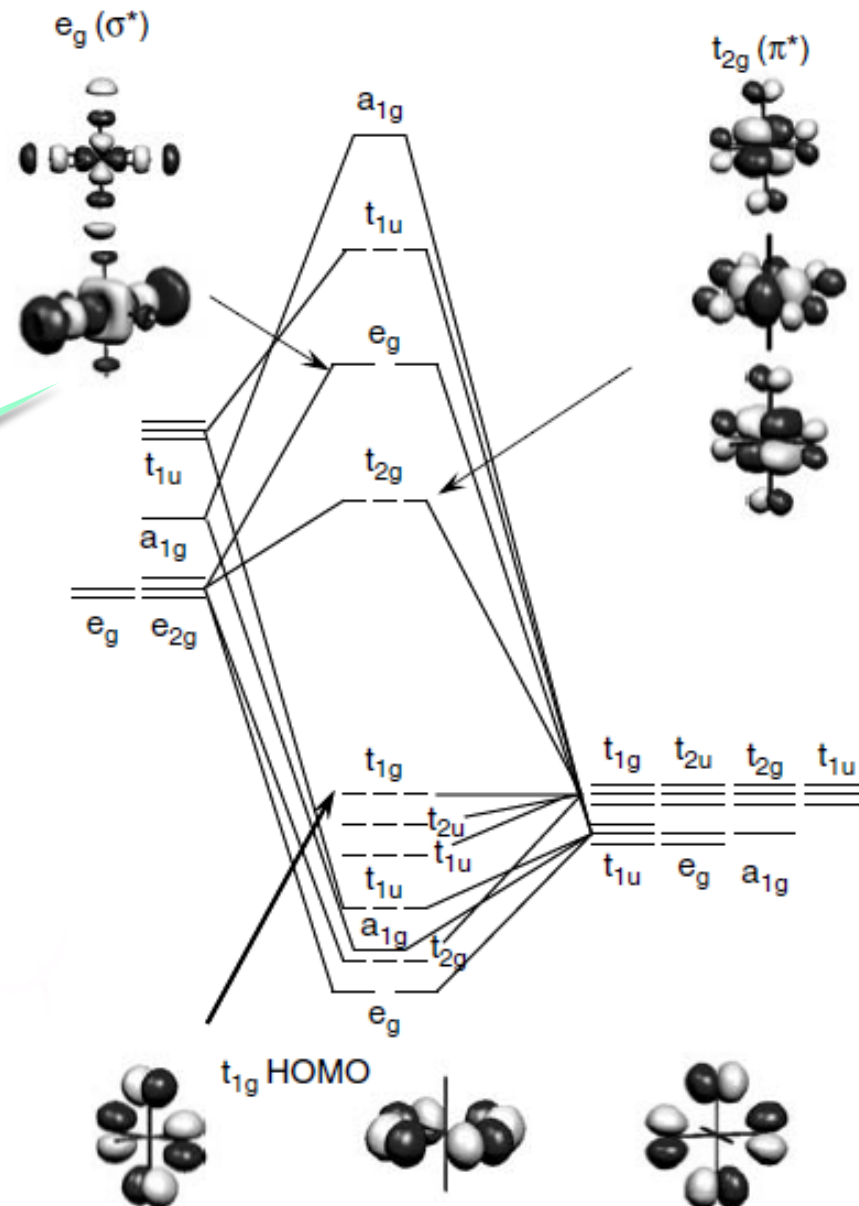
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 12. **Summary and Comparison of the LCAO and Valence Bond Methods**

Weaknesses of LCAO-MO (1)

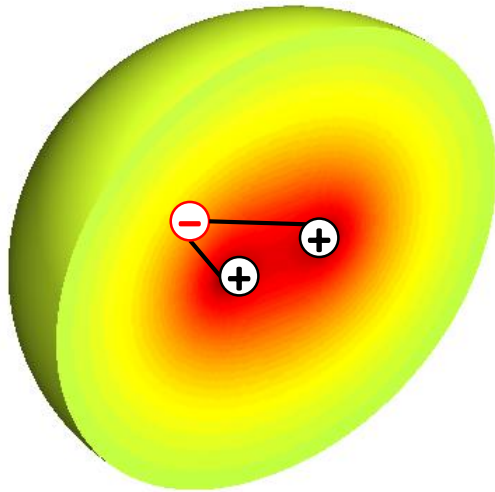
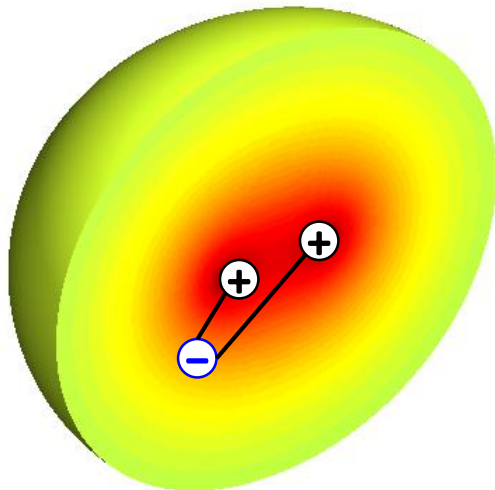
Many atomic orbitals combine to make one molecular orbital.

- Non-intuitive
- Require computer codes

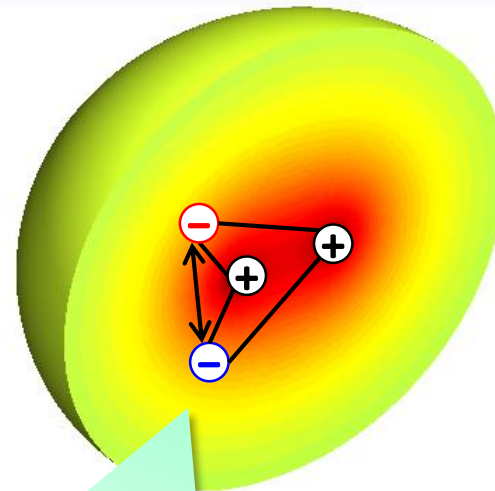
TiO_6^{8-}
(in crystal)



Weaknesses of LCAO-MO (2)

 $\times$ 

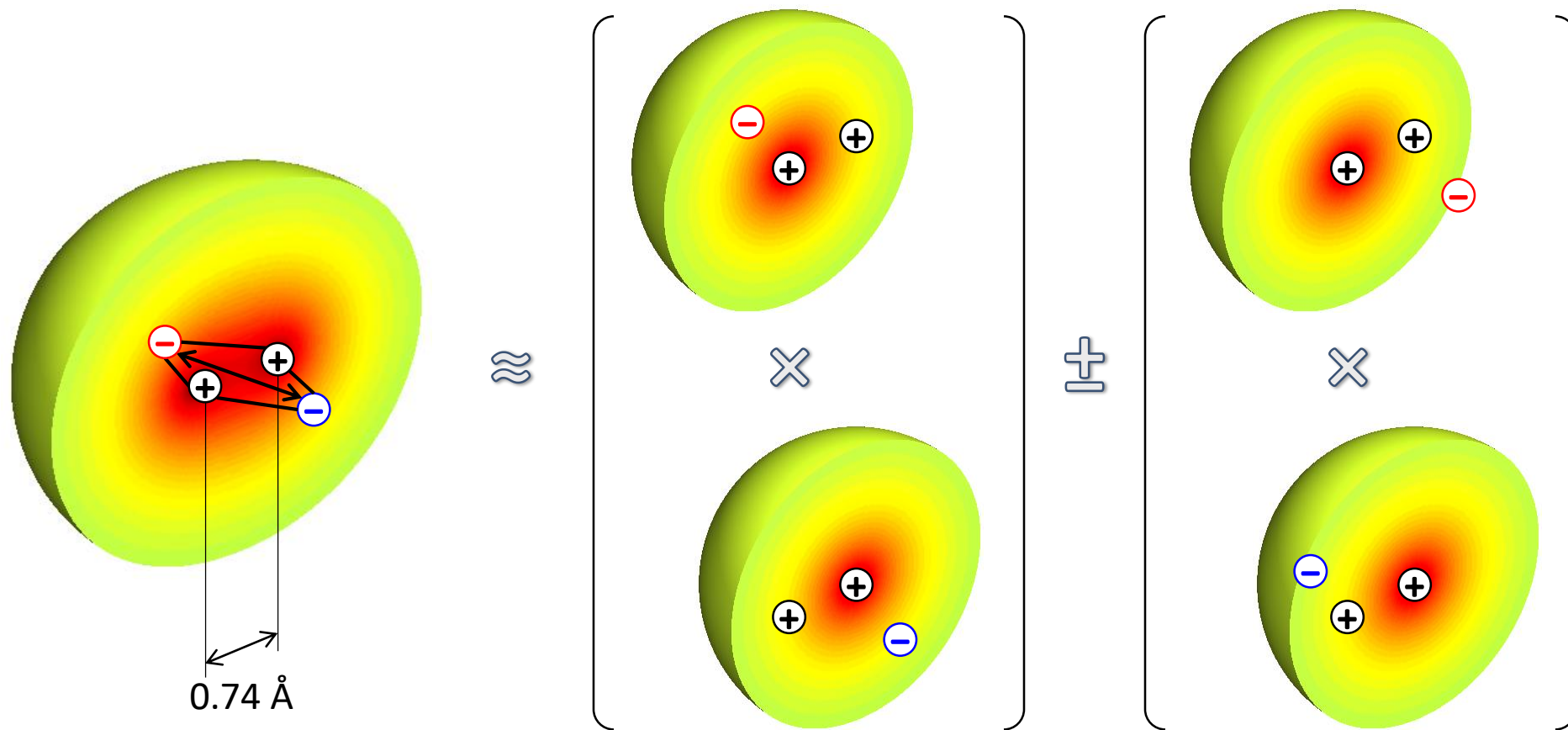
$$\psi(x_1, y_1, z_1, x_2, y_2, z_2) \approx \varphi(x_1, y_1, z_1) \cdot \varphi(x_2, y_2, z_2)$$
$$\Rightarrow \psi(1, 2) \approx \varphi(1) \cdot \varphi(2)$$

 $\approx$ 

$\text{H}^- \text{H}^+$ is very unlikely for H_2 !

The Valence Bond Theory

$\text{H}-\text{H} \approx \text{H}^{\bullet} \times \text{H}$ or $\text{H} \times^{\bullet} \text{H}$, but neither $\text{H} \text{H}^{\bullet} \times$ nor $\times^{\bullet} \text{H} \text{H}$.

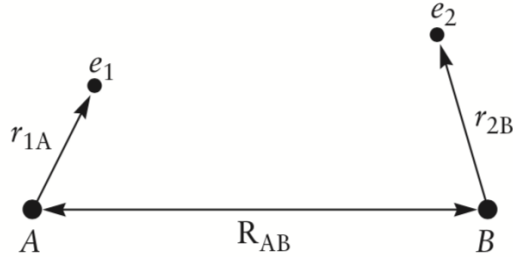


Valence Bond Theory and the Electron Pair Bond

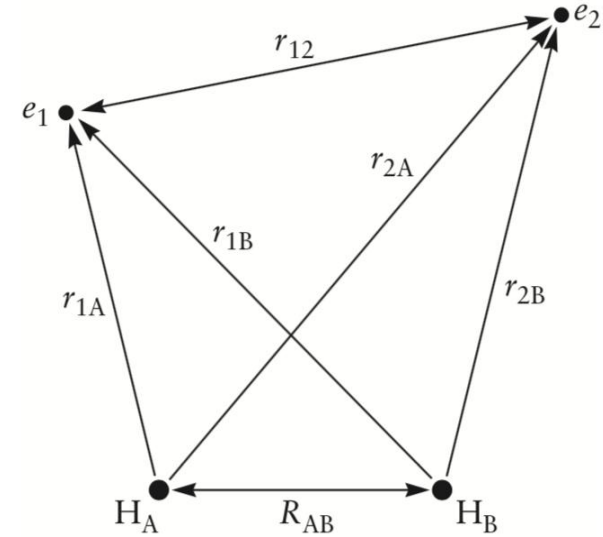
Valence bond (VB) model:

- Grew out of the **qualitative Lewis electron-pair model** in which the chemical bond is described as **a pair of electrons that is localized between two atoms**.
- The VB model **constructs a wave function for the bond** by assuming that **each atom arrives with at least one unpaired electron in an AO**.
- The resulting wave function: **a product of two one-electron wave functions**, each describing an electron localized on one of the arriving atoms. (The spins of the electrons must be paired to satisfy the Pauli exclusion principle.)

Single Bonds (H-H)



$$\varphi^A(r_{1A})\varphi^B(r_{2B})$$



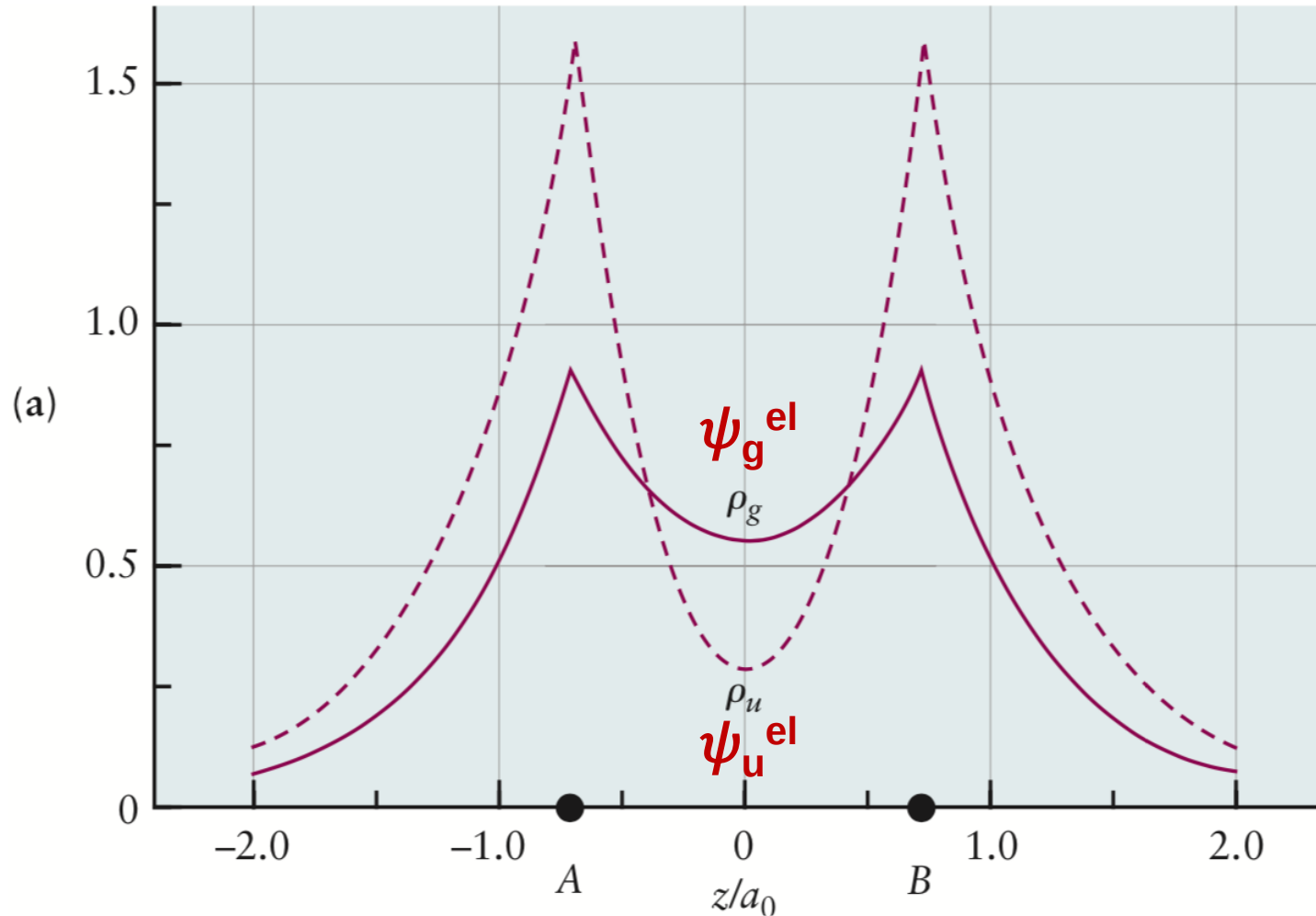
$$\psi^{\text{el}}(r_{1A}, r_{2B}; R_{AB}) = c(R_{AB})\varphi^A(r_{1A})\varphi^B(r_{2B})$$

The electrons are indistinguishable: as the atoms begin to interact strongly, we cannot determine whether electron 1 arrived with proton A and electron 2 with proton B, or vice versa.

$$\psi^{\text{el}}(r_{1A}, r_{2B}; R_{AB}) = c_1(R_{AB})\varphi^A(r_{1A})\varphi^B(r_{2B}) + c_2(R_{AB})\varphi^A(r_{2A})\varphi^B(r_{1B}) \quad c_1 = c_2 \text{ and } c_1 = -c_2$$

Symmetry requires that $c_1 = c_2$ and $c_1 = -c_2$ are equally valid choices

Single Bonds (H-H)



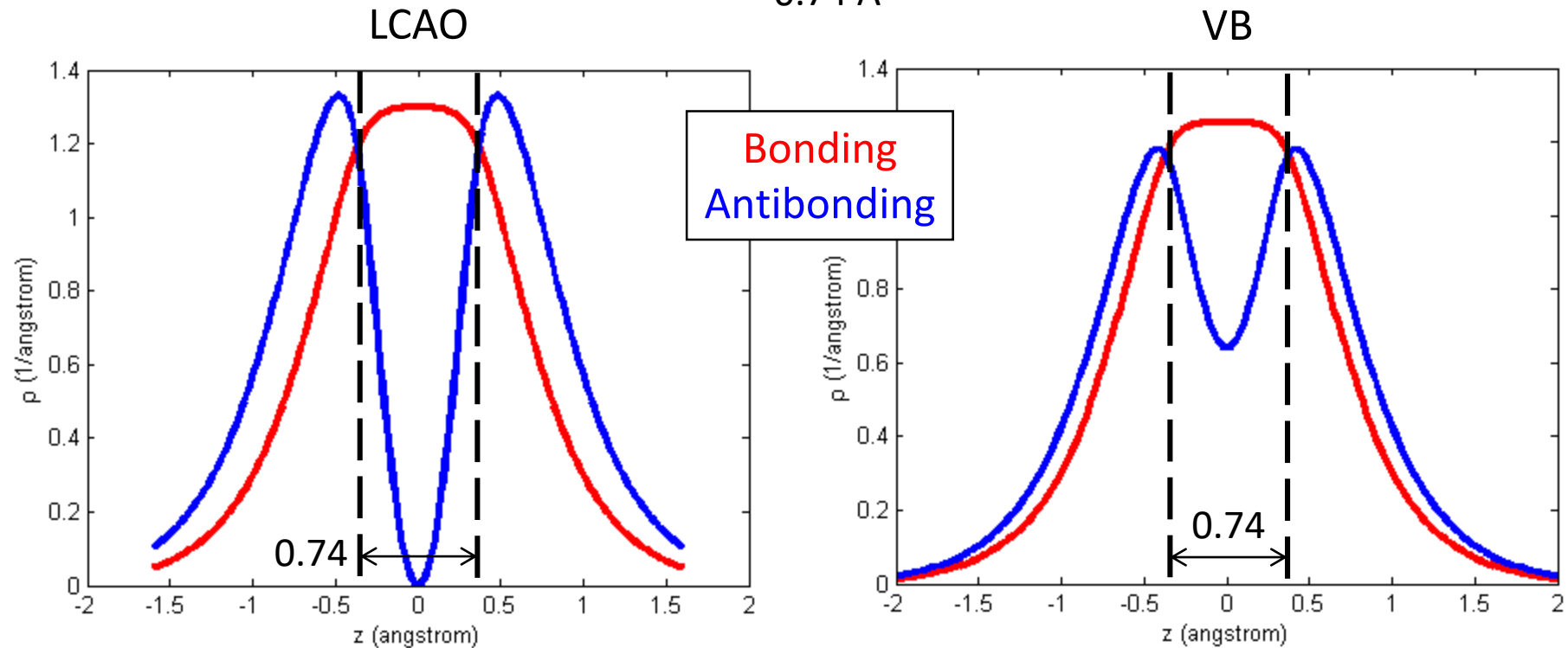
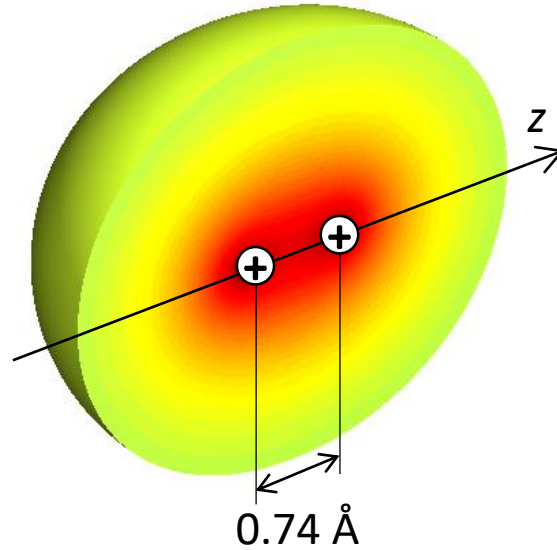
$$\psi_g^{\text{el}} = C_1[1s^{\text{A}}(1)1s^{\text{B}}(2) + 1s^{\text{A}}(2)1s^{\text{B}}(1)]$$



FIGURE 6.24 The electron density for the ψ_g^{el} and ψ_u^{el} wave functions in the simple valence bond model for H_2 . (a) The electron density ρ_g for ψ_g^{el} and ρ_u for ψ_u^{el} calculated analytically as described in the text. (b) Three-dimensional isosurface of the electron density for the ψ_g^{el} wave function obtained by computer calculations.

(Courtesy of Mr. Hatem Helal and Professor William A. Goddard III, California Institute of Technology, and Dr. Kelly P. Gaither, University of Texas at Austin.)

Comparison between LCAO & VB



Characteristics of the two theories

LCAO

- Ignores e-e repulsion
- Electrons wander around
- Orbitals: delocalized (离域)

VB

- Exaggerates e-e repulsion
- Electrons adhere to the nuclei
- Orbitals: localized (定域/局域)

Pros and Cons of VB Theory



Works best for **localized bonds**

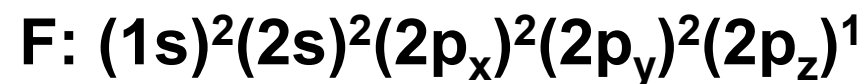
Difficult to estimate **energy levels**



More accurate than LCAO for σ bonds

Compatible with **hybridization 杂化**

Single Bonds (F-F)



Lewis diagram for F₂.

$$\psi_g^{\text{bond}} = C_1[2p_z^{\text{A}}(1)2p_z^{\text{B}}(2) + 2p_z^{\text{A}}(2)2p_z^{\text{B}}(1)]$$

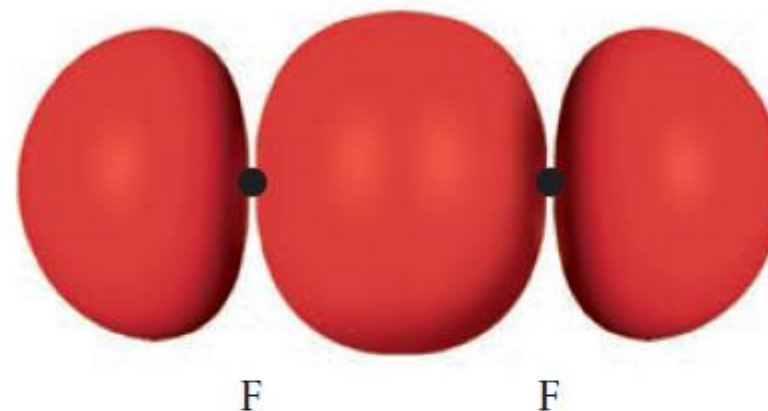


FIGURE 6.25 Isosurface representation of the electron density in the F₂ σ bond formed from a pair of electrons initially localized in a $2p_z$ orbital on each F atom. (Courtesy of Mr. Hatem Helal and Professor William A. Goddard III, California Institute of Technology, and Dr. Kelly P. Gaither, University of Texas at Austin.)

Single Bonds (H-F)

HF: one shared pair in a single bond produced by overlap of H(1s) and F(2p_z).

$$\psi^{\text{bond}} = C_1[2p_z^{\text{F}}(1)1s^{\text{H}}(2)] + C_2[2p_z^{\text{F}}(2)1s^{\text{H}}(1)]$$

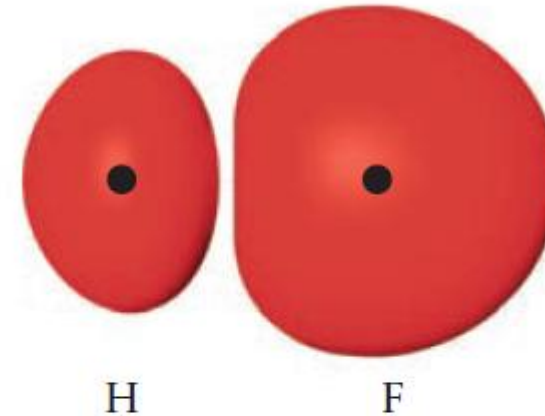


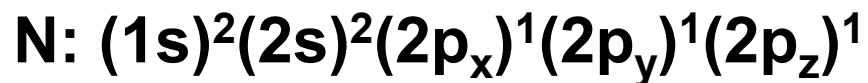
FIGURE 6.26 Isosurface representation of the electron density in the HF σ bond formed from a pair of electrons initially localized in a 1s orbital on H and in a 2p_z orbital on F. (Courtesy of Mr. Hatem Helal and Professor William A. Goddard III, California Institute of Technology, and Dr. Kelly P. Gaither, University of Texas at Austin.)

Single Bonds

These wave functions should **not be called MOs**: they are not single-electron wave functions and they are **not de-localized over the entire molecule**.

Each bond is formed by **overlap of two AOs**, each of which is already half filled with one electron. The electrons in the two participating AOs must have **opposite spin**. (the bond corresponds to an electron pair with opposite, or “paired,” spins)

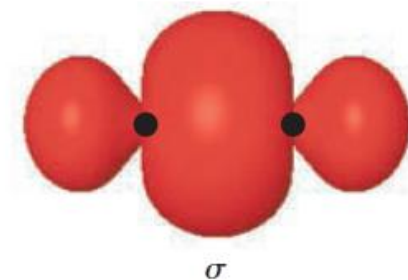
Multiple Bonds (N₂)



Lewis diagram for N₂.

- When approach one another along the z-axis (2p_z: head-on)

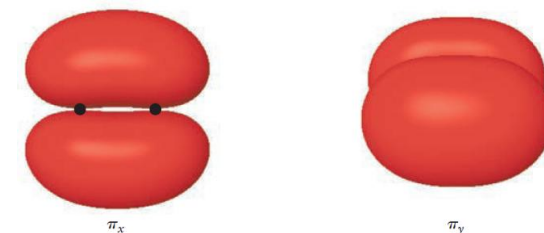
$$\psi_{\sigma}^{\text{bond}} = C_1[2p_z^A(1)2p_z^B(2) + 2p_z^A(2)2p_z^B(1)]$$



- The 2p_x and 2p_y orbitals: side-to-side.

$$\psi_{\pi_x}^{\text{bond}}(1, 2; R_{AB}) = C_1(R_{AB})[2p_x^A(1)2p_x^B(2)] + C_1(R_{AB})[2p_x^A(2)2p_x^B(1)]$$

$$\psi_{\pi_y}^{\text{bond}}(1, 2) = C_1[2p_y^A(1)2p_y^B(2) + 2p_y^A(2)2p_y^B(1)]$$

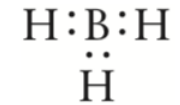
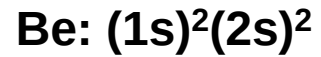


Polyatomic Molecules

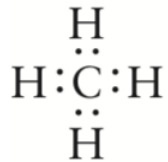
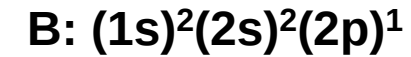
Valence Bond Model vs. VSEPR Model



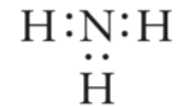
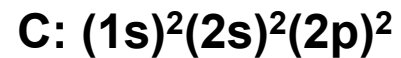
Lewis diagram for BeH_2 .



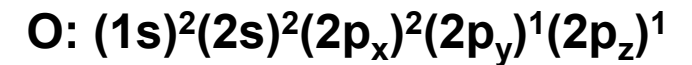
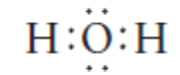
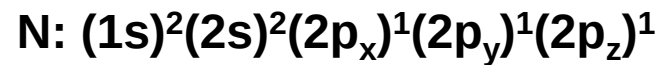
Lewis diagram for BH_3 .

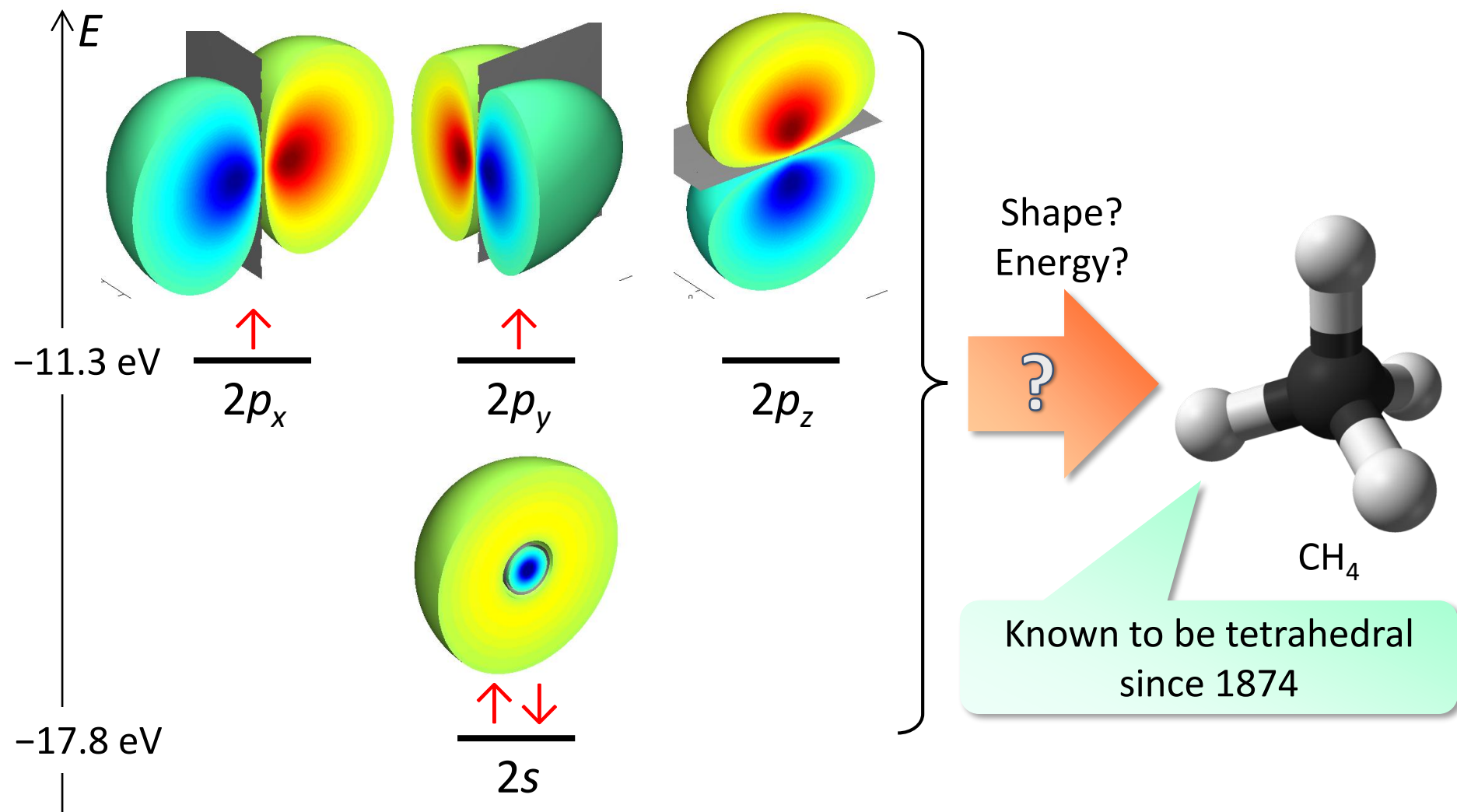


Lewis diagram for CH_4 .



Lewis diagram for NH_3 .

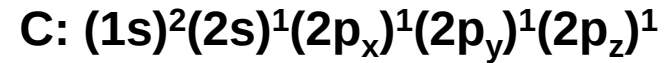
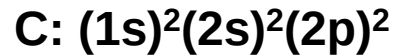
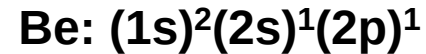
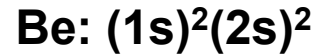




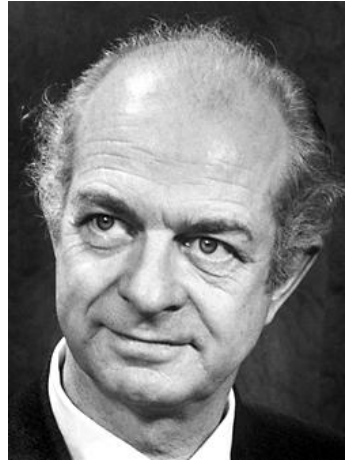
Polyatomic Molecules

Linus Pauling gave the answer in 1931: the concepts of **promotion (激发)** and **hybridization (杂化)**.

Promotion of valence electrons **from the ground state to excited states** at higher energy

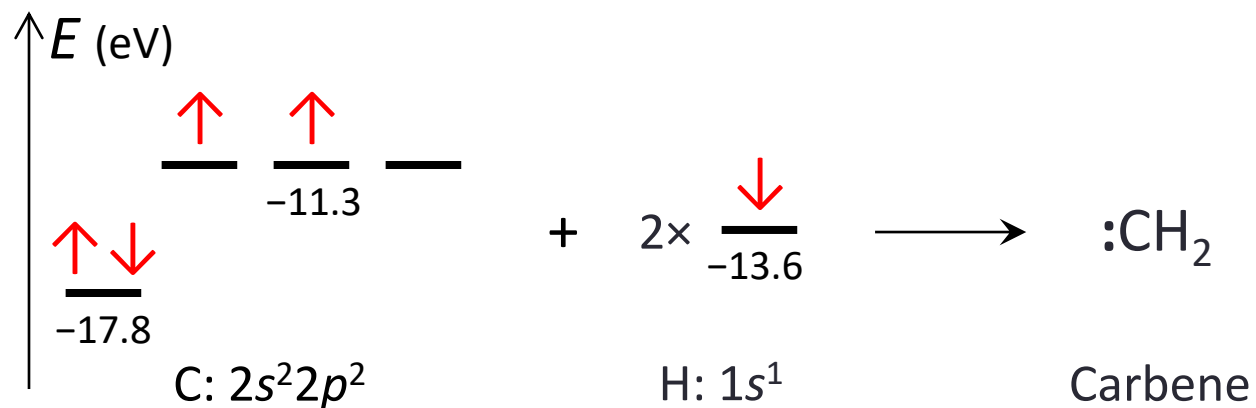


Hybridization of 2s and 2p orbitals after promotion: form new orbitals with the **proper symmetry** for bond formation

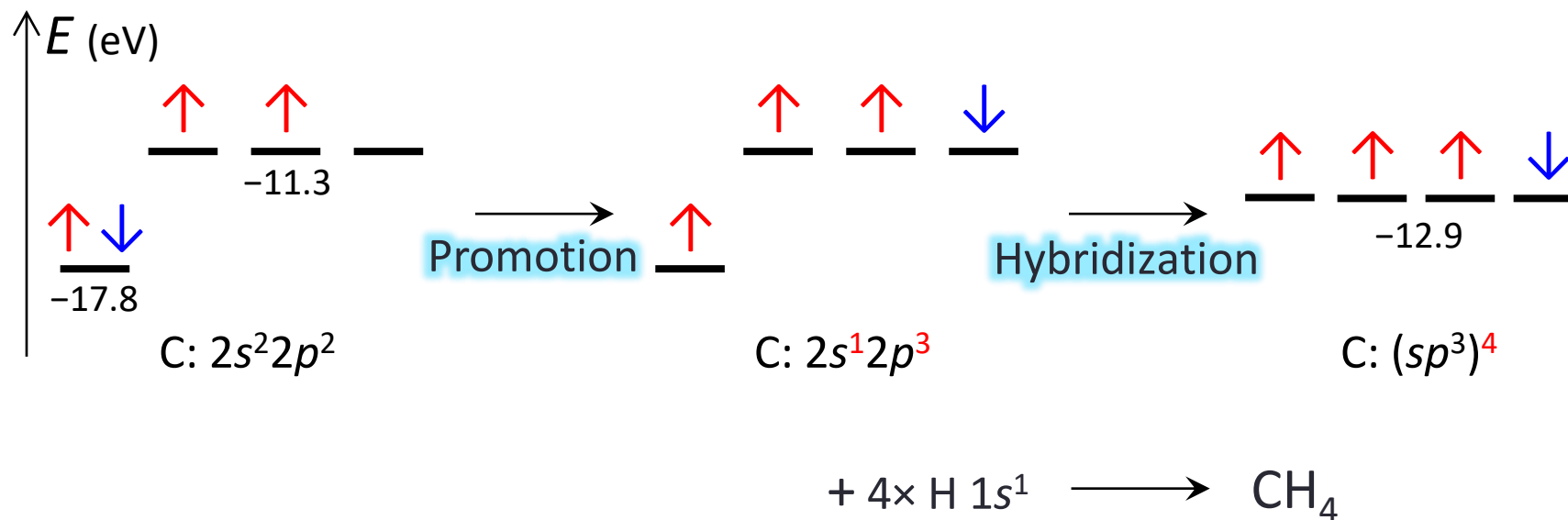


Linus Pauling
(1901–1994)

Pauling's Answer



Or better,



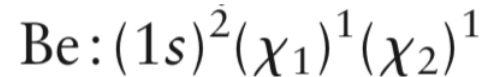
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SP hybridization

BeH₂: linear

sp hybrid atomic orbitals: formed as the **sum or difference** of **one s orbital and one p orbital**.



lowercase Greek letter chi, χ , to represent hybrid orbitals

$$\chi_1(r) = \frac{1}{\sqrt{2}} [2s + 2p_z]$$

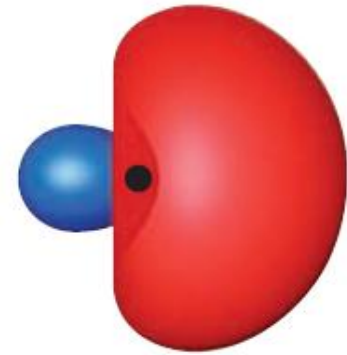
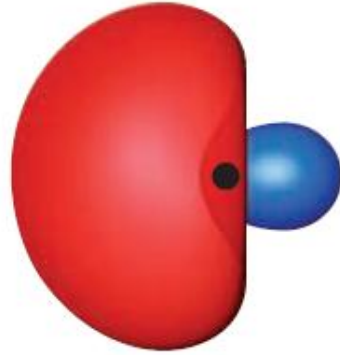
$$\chi_2(r) = \frac{1}{\sqrt{2}} [2s - 2p_z]$$

The coefficient $1/\sqrt{2}$ is a normalization constant

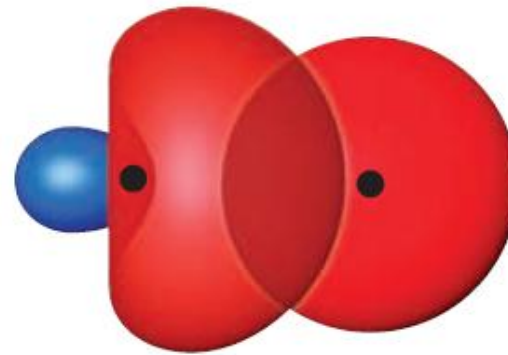
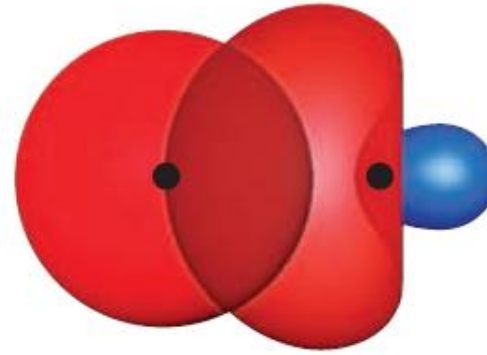
SP hybridization



$2s$



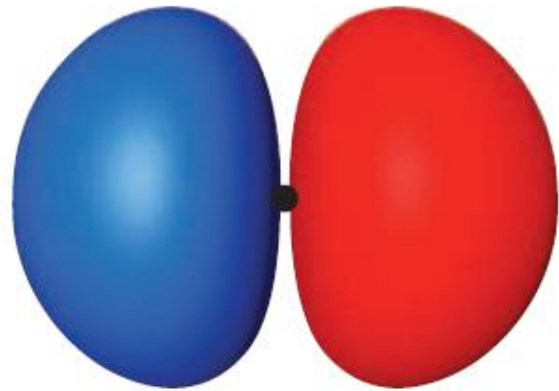
sp hybrid orbitals



Hybrid orbital overlap



Electron density



$2p_z$

Atomic orbitals

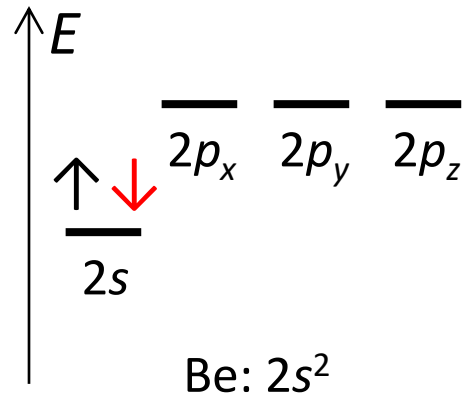
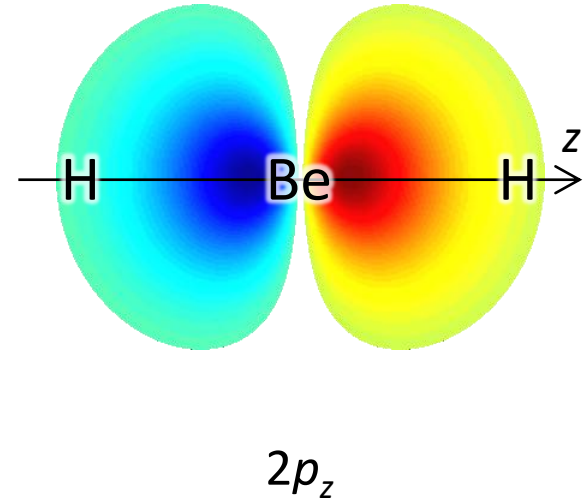
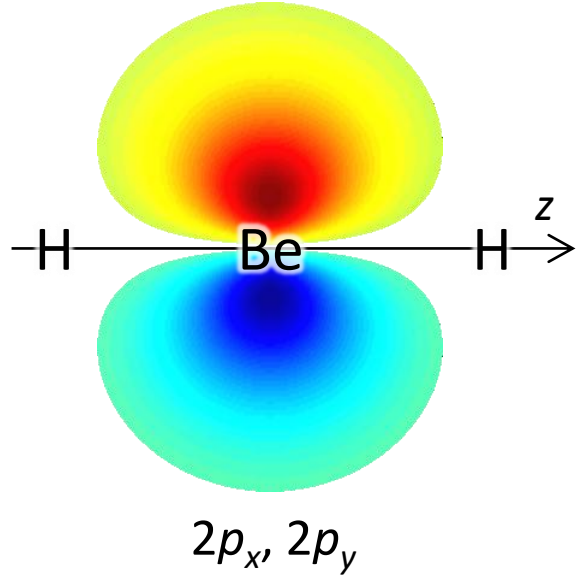
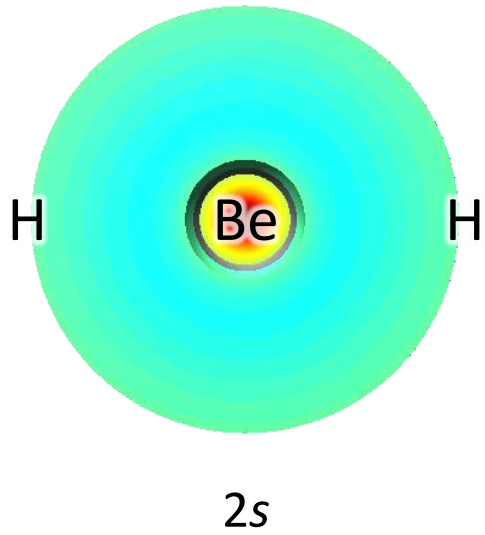
(a)

(b)

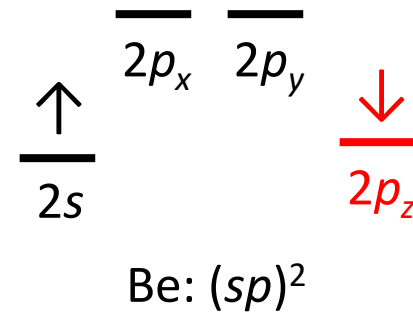
(c)

(d)

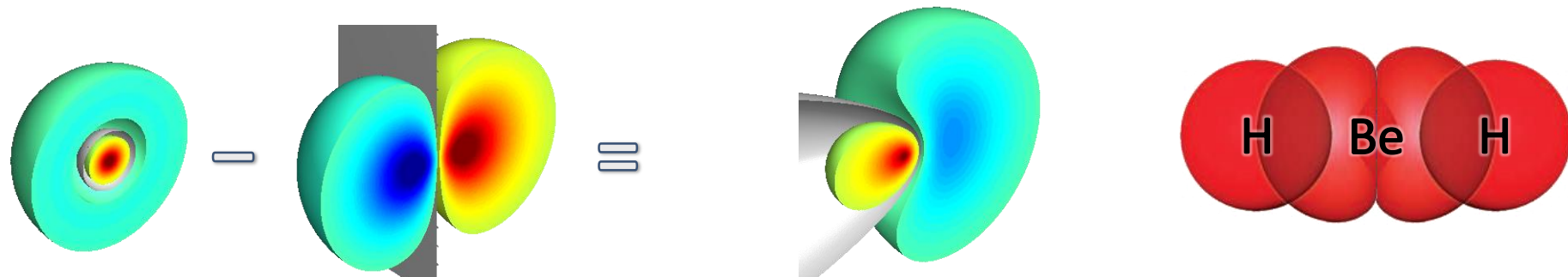
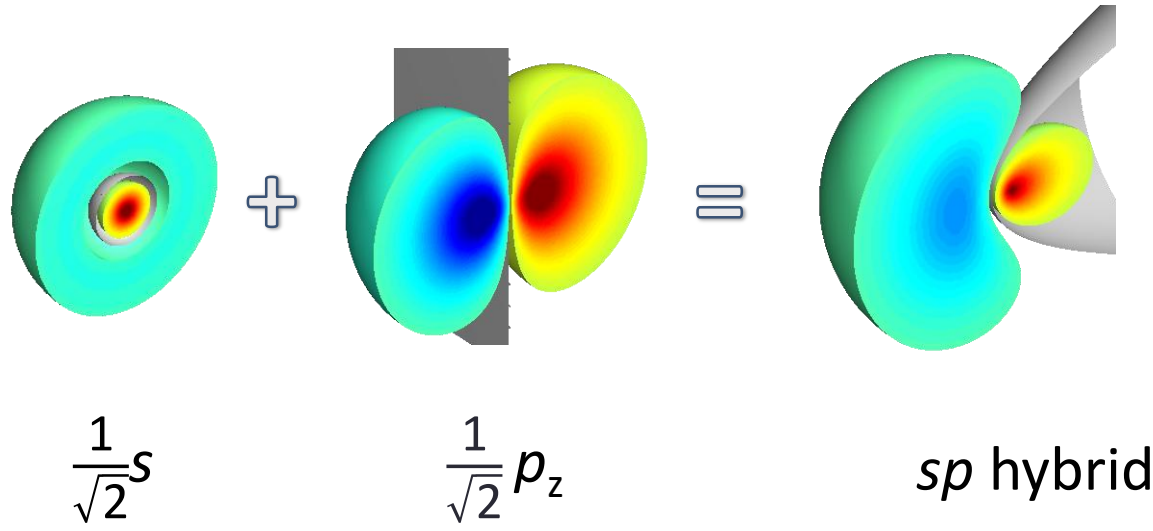
Why Hybridize? BeH₂



External field



SP hybridization



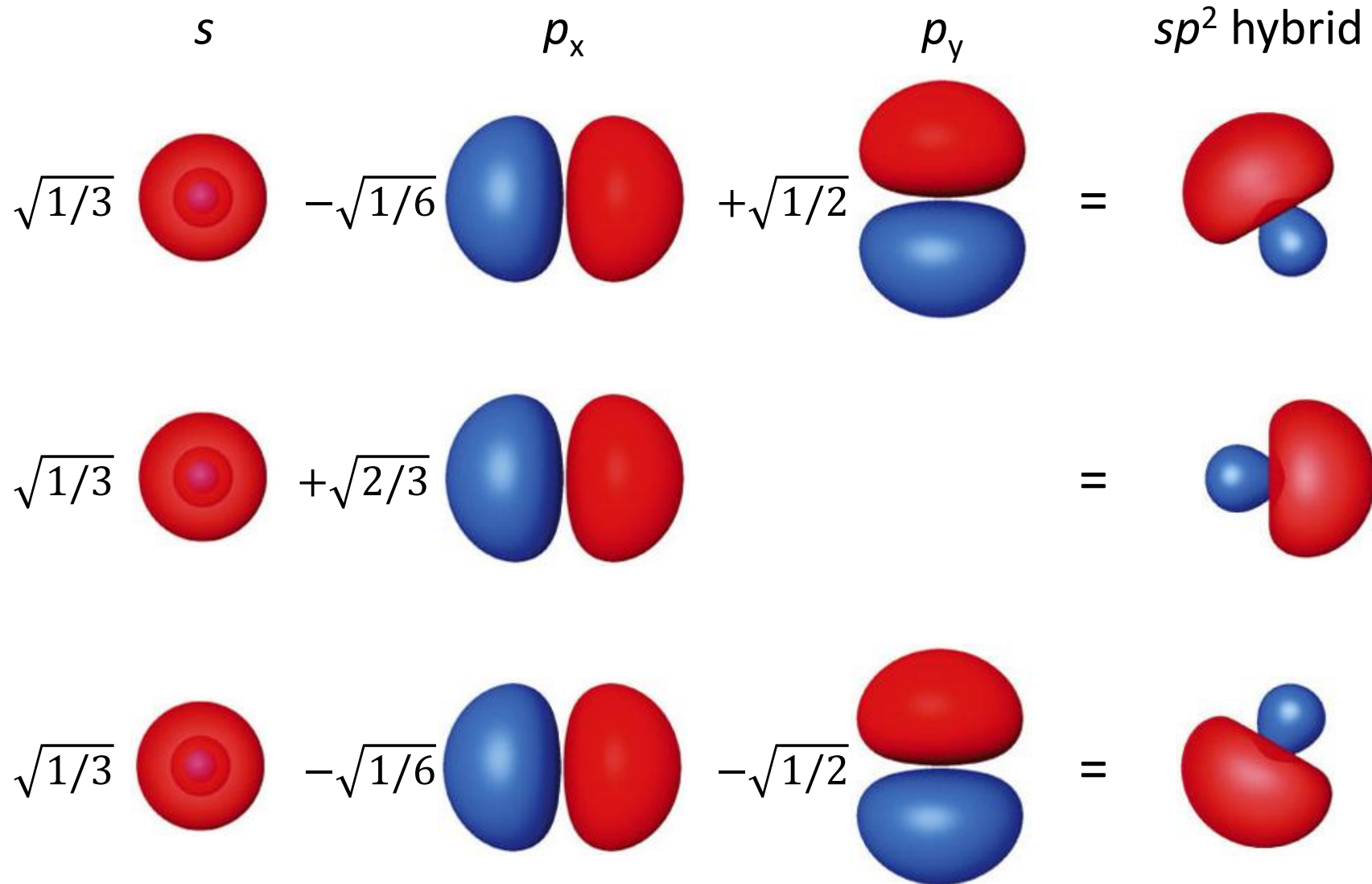
SP² hybridization

BH₃: a trigonal planar structure with three equivalent bonds

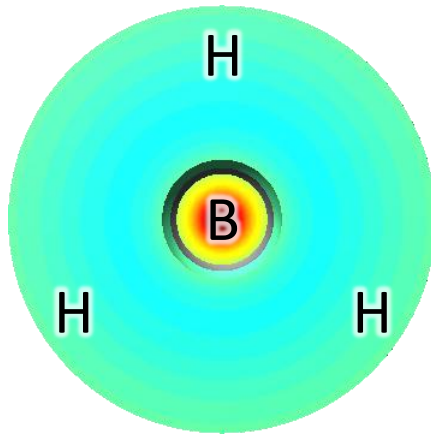
Promotion of one of the 2s electrons creates the excited-state configuration **B: (1s)² (2s)¹(2p_x)¹(2p_y)¹**, and these three AOs can be mixed to form the three equivalent new orbitals:

sp² hybrid atomic orbitals: formed from **one s orbital and two p orbitals**.

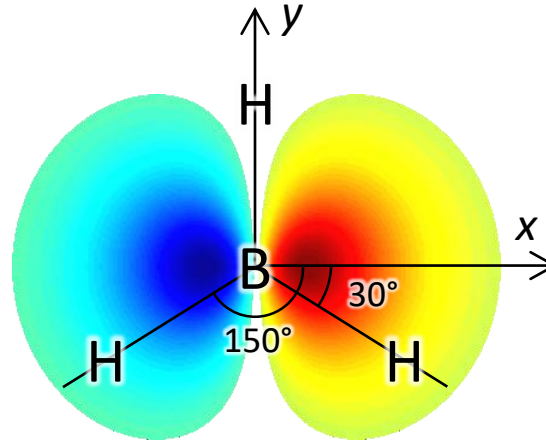
SP² hybridization



Why Hybridize? BH_3



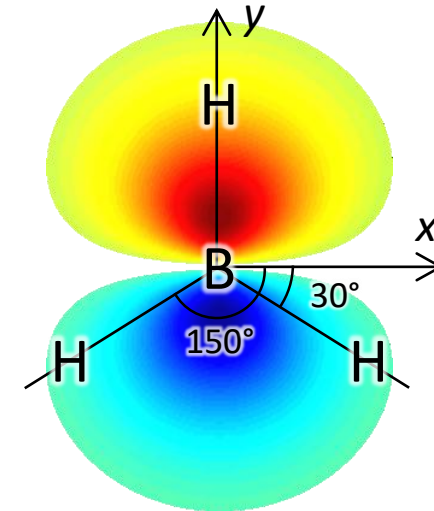
$2s$



$$2p_x$$

$$\cos^2(30^\circ) + \cos^2(150^\circ)$$

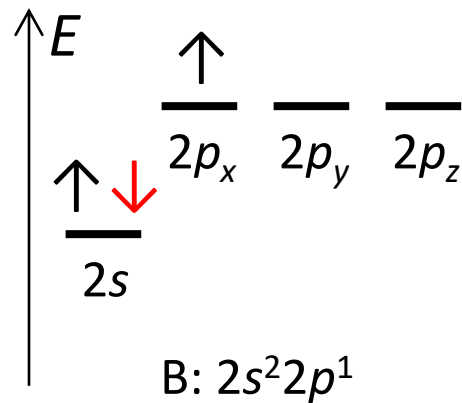
$$= \frac{3}{2}$$



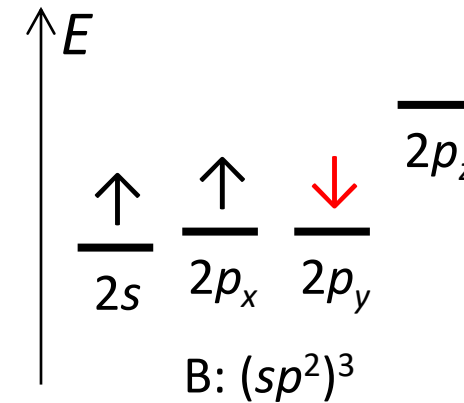
$$2p_y$$

$$\sin^2(30^\circ) + \sin^2(150^\circ)$$

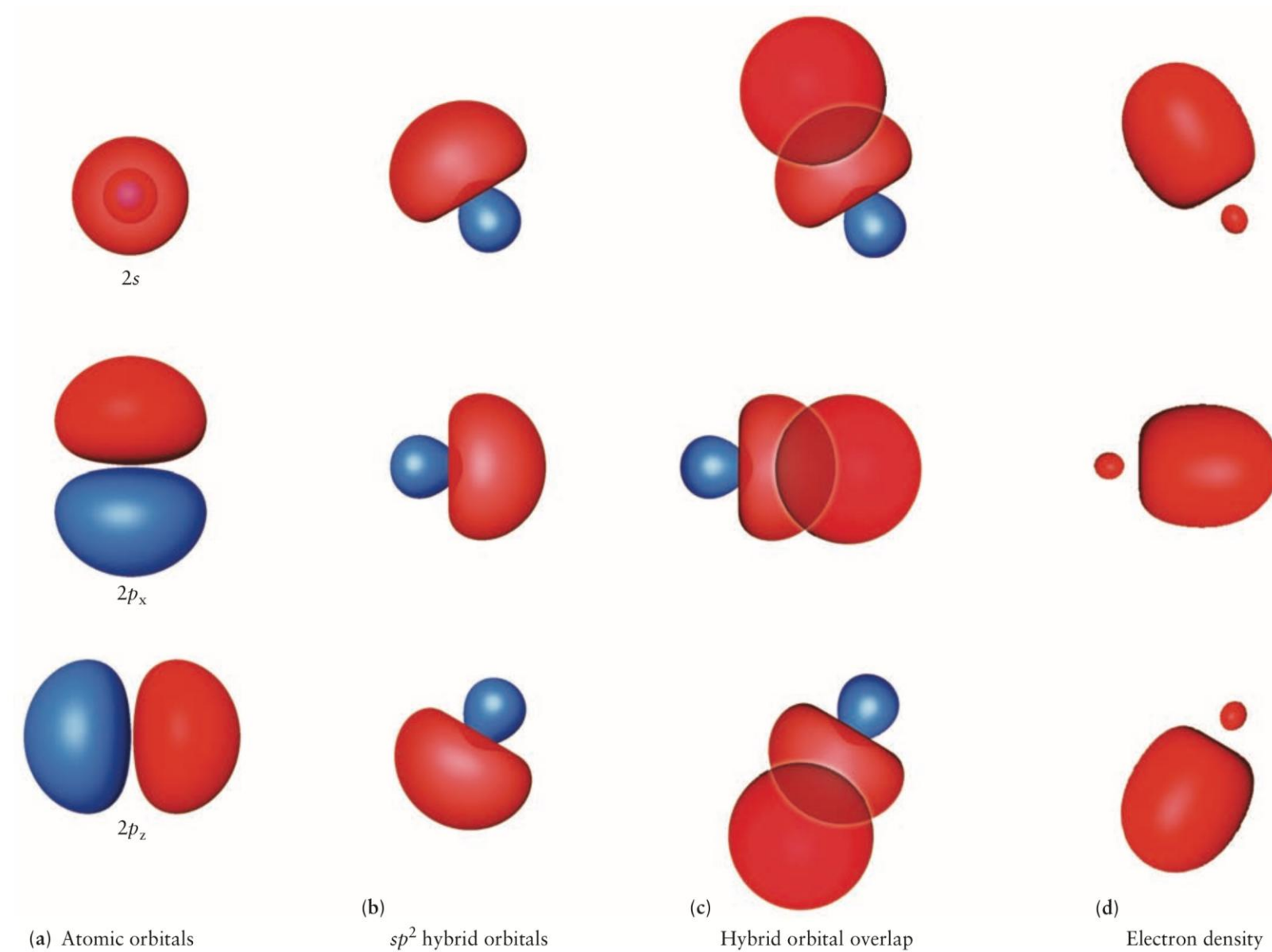
$$+ \sin^2(-90^\circ) = \frac{3}{2}$$



External field



SP² hybridization



SP³ hybridization

CH₄: a tetrahedral structure with four equivalent bonds

Promotion of one of the 2s electrons creates the excited-state configuration **C: (1s)² (2s)¹(2p_x)¹(2p_y)¹(2p_z)¹**, and these four AOs can be mixed to form the four equivalent new orbitals:

$$\chi_1(r) = \frac{1}{2} [2s + 2p_x + 2p_y + 2p_z]$$

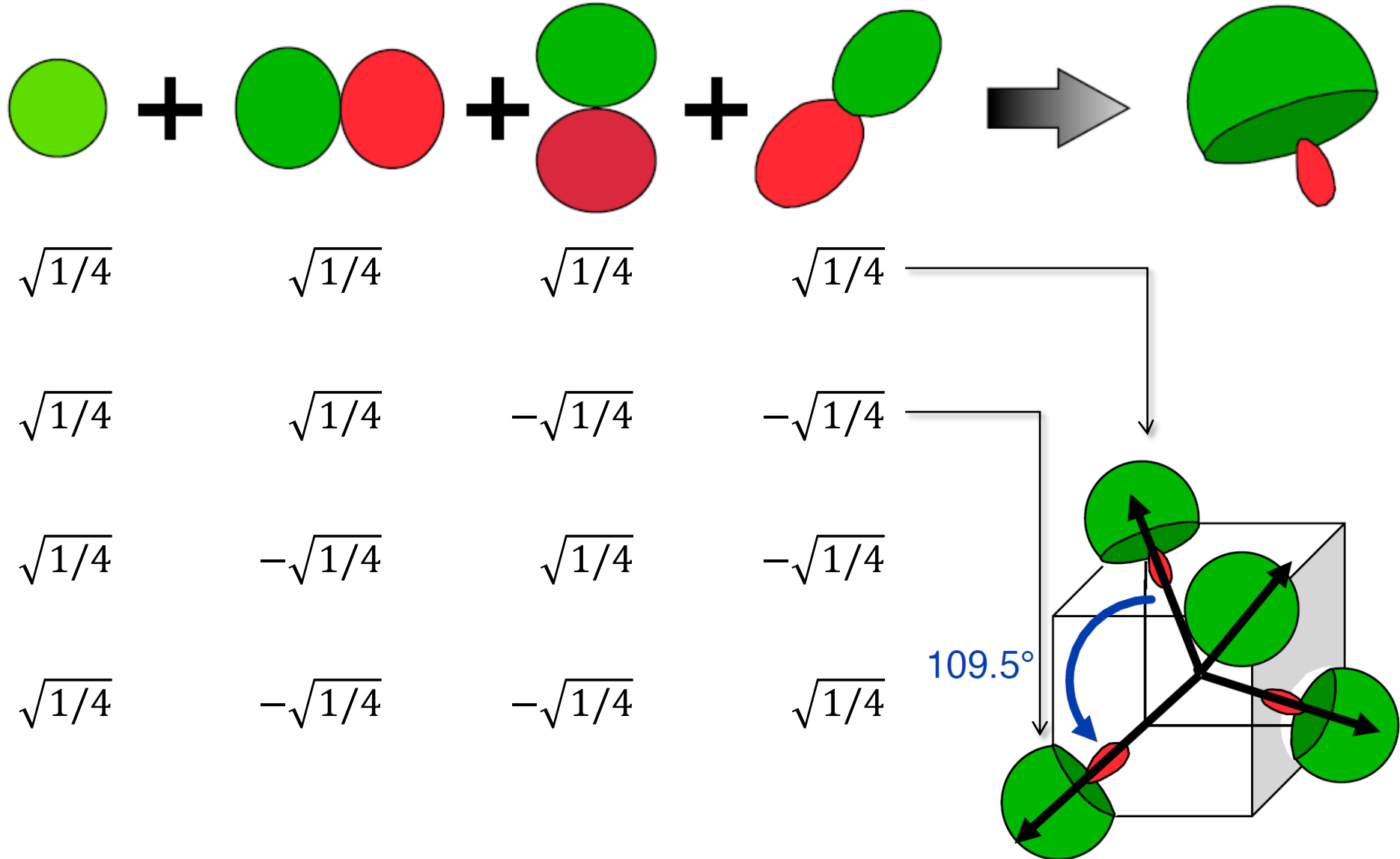
$$\chi_2(r) = \frac{1}{2} [2s - 2p_x - 2p_y + 2p_z]$$

$$\chi_3(r) = \frac{1}{2} [2s + 2p_x - 2p_y - 2p_z]$$

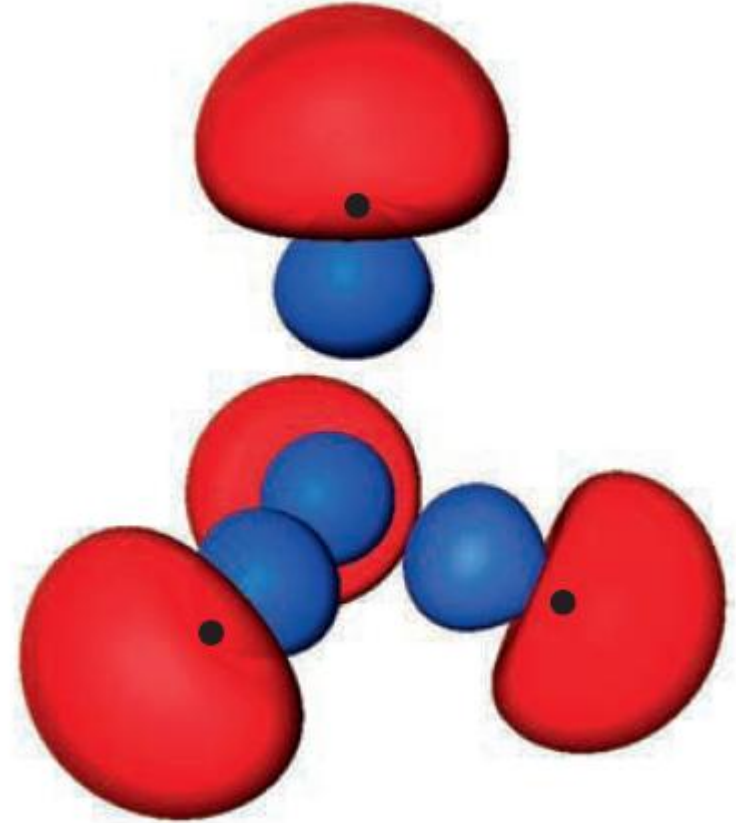
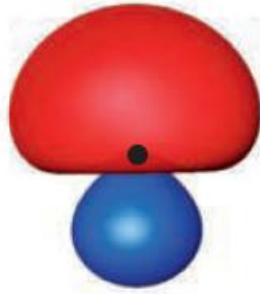
$$\chi_4(r) = \frac{1}{2} [2s - 2p_x + 2p_y - 2p_z]$$

sp³ hybrid atomic orbitals: formed from **one s orbital and three p orbitals**.

SP³ hybridization



SP³ hybridization

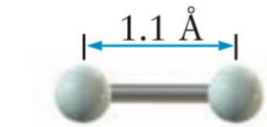


Major points of today's lecture

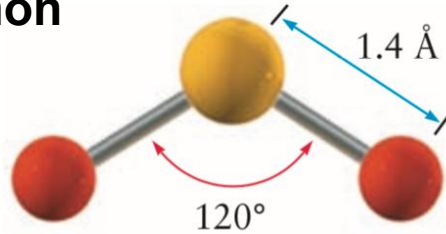
- **Atomic shells and classical models of chemical bonding**
 1. The shapes of molecules
 2. Oxidation numbers
 3. Inorganic nomenclature

The shapes of molecules

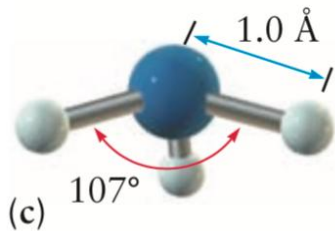
- **Bond length:** distance between the atomic nuclei in a particular bond
- **Bond angle:** angle between the axes of adjacent bonds
- **Dihedral angle:** angle between two planes, each defined by three atoms with the planes having two atoms in common



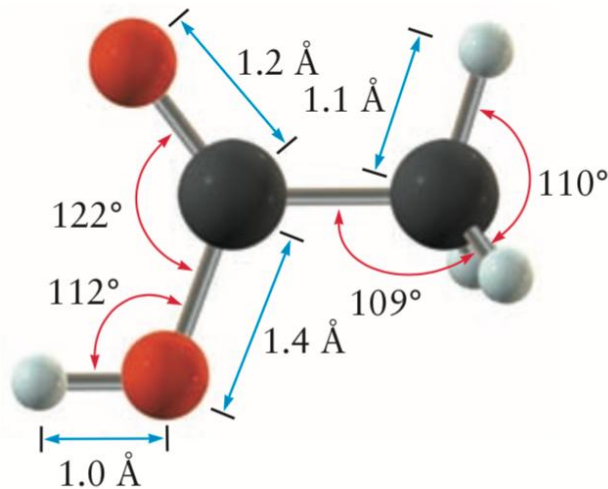
(a)



(b)



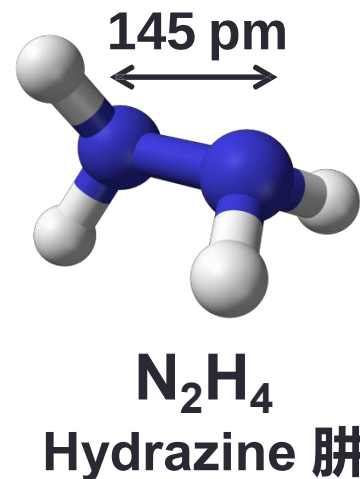
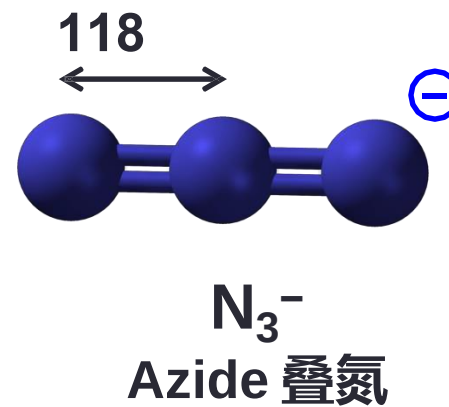
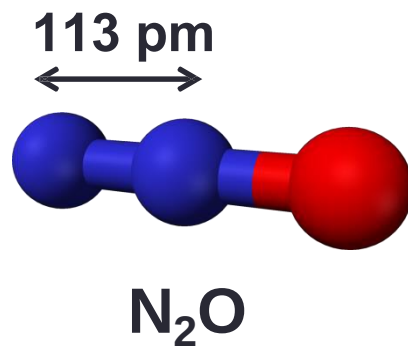
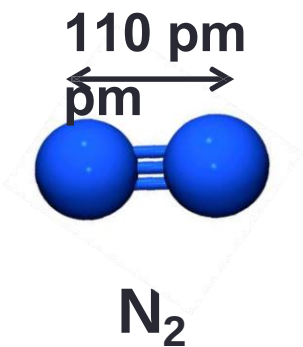
(c)



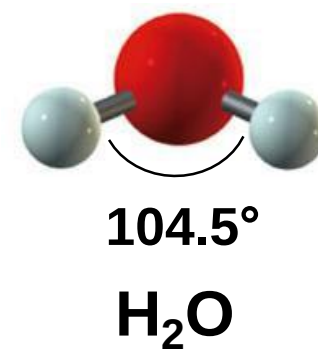
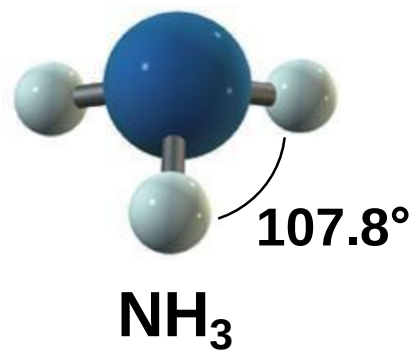
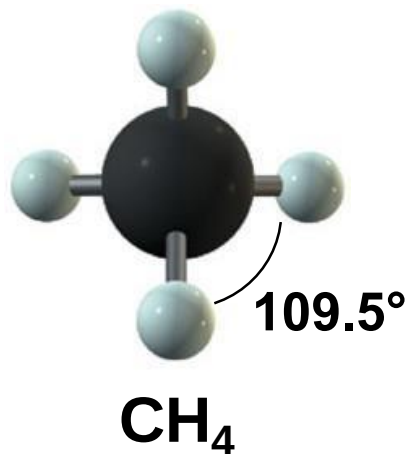
(d)

Molecular shape or geometry is governed by energetics; a molecule assumes the geometry that gives it the **lowest potential energy**.

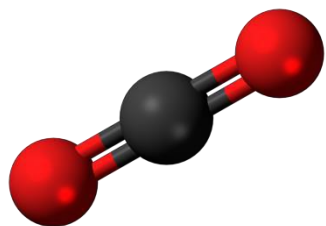
Polyatomic molecules: bond lengths



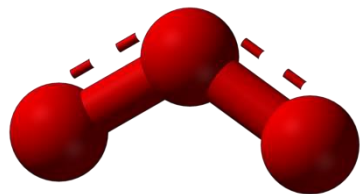
Polyatomic molecules: bond angles



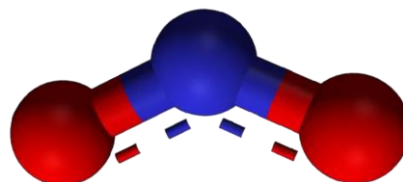
Polyatomic molecules: bond angles



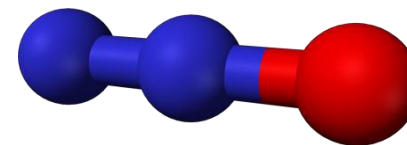
CO₂



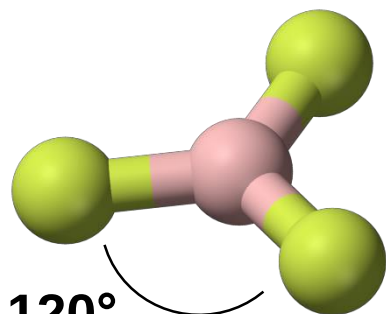
O₃



NO₂

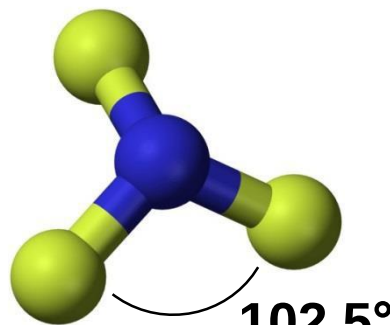


N₂O



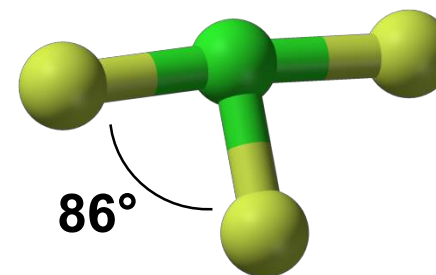
120°

BF₃



102.5°

NF₃



86°

BrF₃

The Valence Shell Electron-Pair Repulsion Theory (VSEPR)

- The Valence Shell Electron-Pair Repulsion Theory (VSEPR): 价层电子对互斥模型



- fundamental idea: the electron pairs position themselves as far apart as possible to minimize their repulsions



$$SN = \left(\begin{array}{c} \text{number of atoms} \\ \text{bonded to central atom} \end{array} \right) + \left(\begin{array}{c} \text{number of lone pairs} \\ \text{on central atom} \end{array} \right)$$

SN: steric number (空间位数)



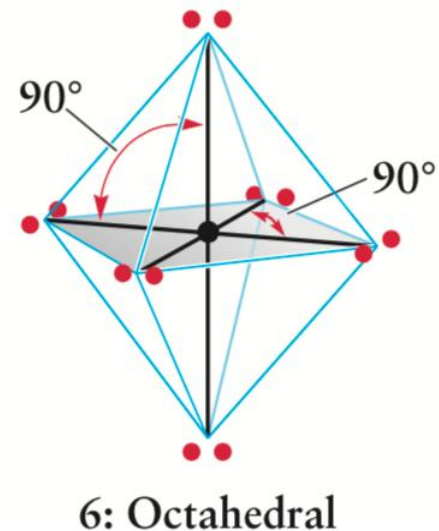
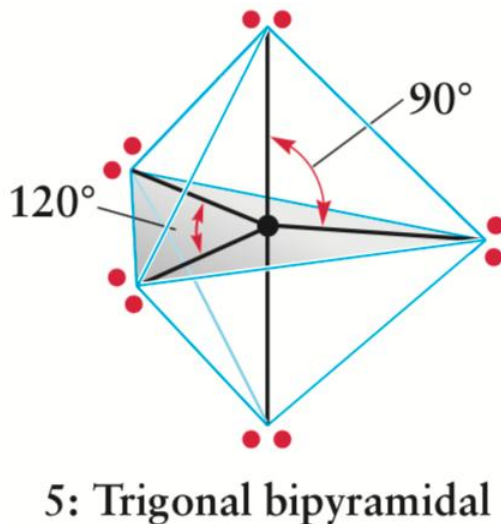
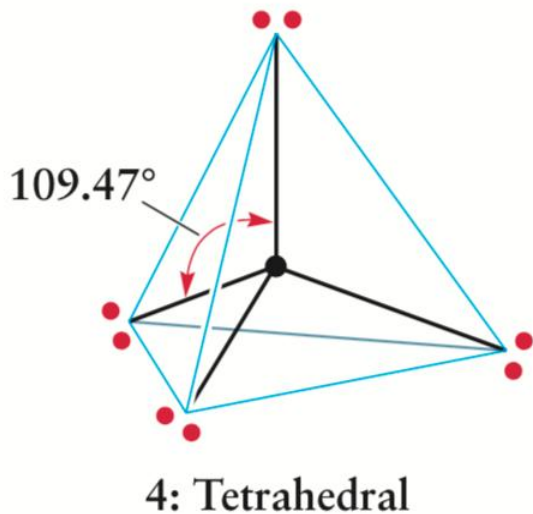
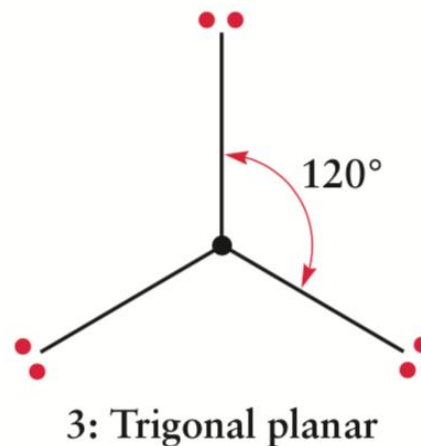
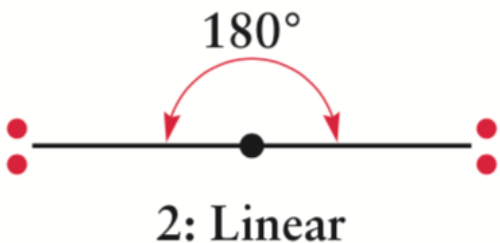
Ronald J. Gillespie
(1924–)



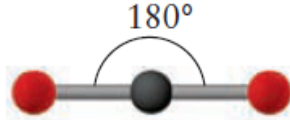
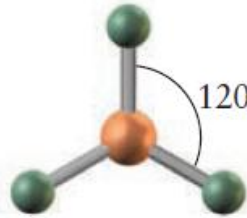
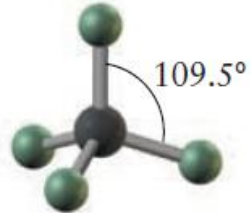
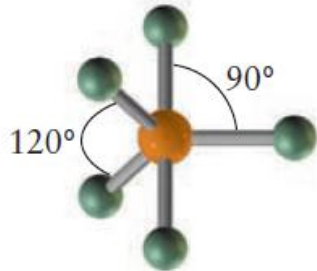
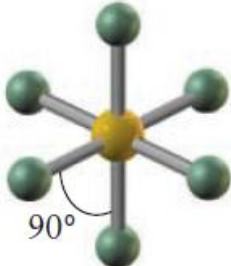
Ronald S. Nyholm
(1917–1971)

The Valence Shell Electron-Pair Repulsion Theory (VSEPR)

SN: steric number (空间位数)



Minimum energy for symmetrical molecules

AX_2	2	Linear		CO_2
AX_3	3	Trigonal planar		BF_3
AX_4	4	Tetrahedral		CF_4
AX_5	5	Trigonal bipyramidal		PF_5
AX_6	6	Octahedral		SF_6

The Valence Shell Electron-Pair Repulsion Theory (VSEPR)

Step 1: Rewrite a structure as AX_mE_n , m = number of ligands, n = number of lone pairs

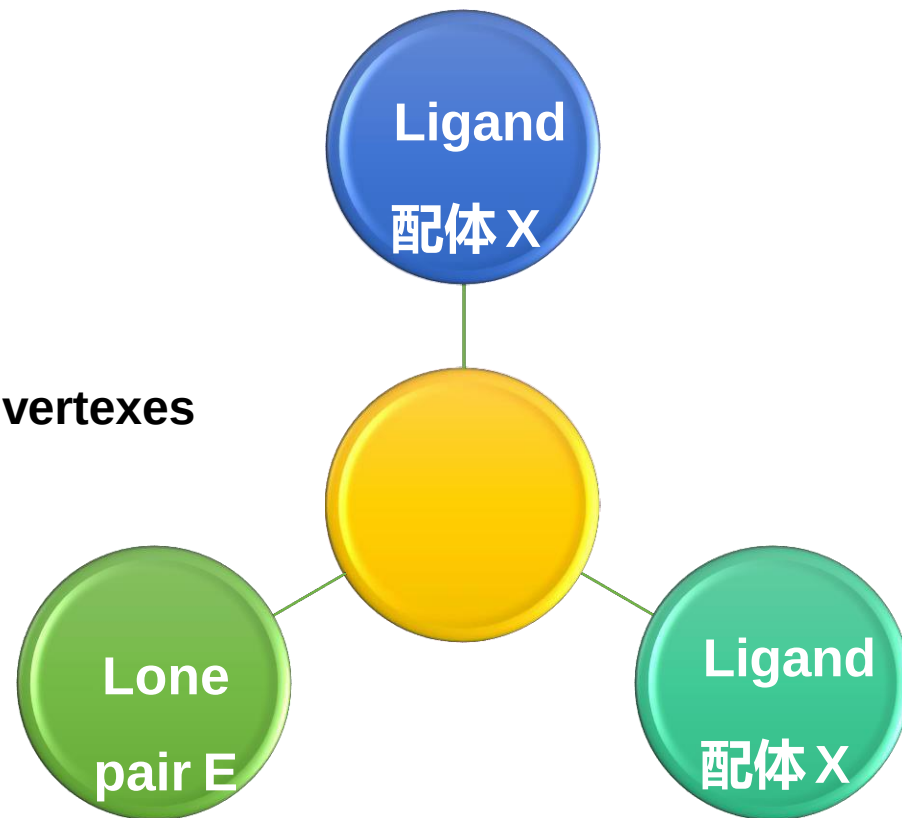
Step 2: Assign a Steric Number; $SN = m + n$

Step 3: Place the ligands and lone pairs on a polyhedron of SN vertexes

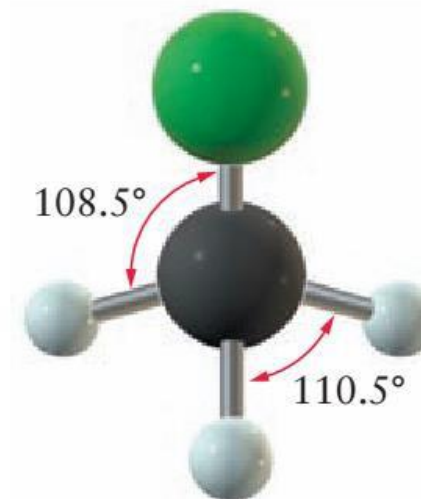
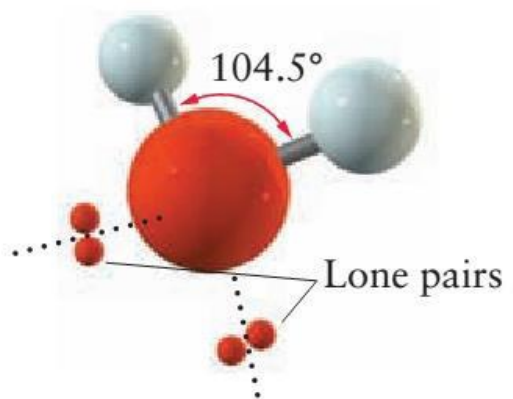
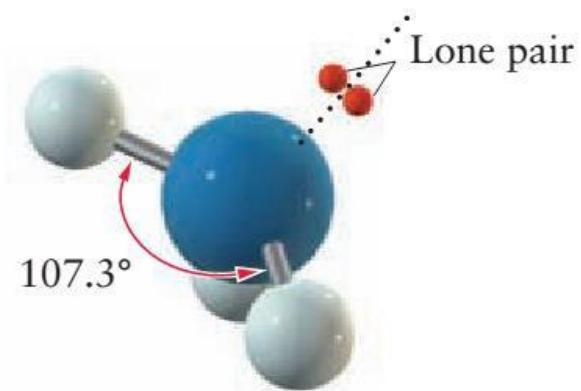
Step 4: Repulsion: $E-E > E-X > X-X$;

triple bond > double bond > single bond

X high electronegativity (EN), less repulsion



Molecular structures

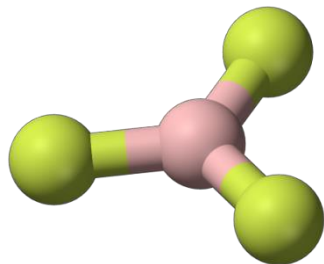


Step 4: Repulsion: $\text{E-E} > \text{E-X} > \text{X-X}$;

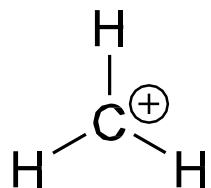
triple bond > double bond > single bond

X high electronegativity (EN), less repulsion

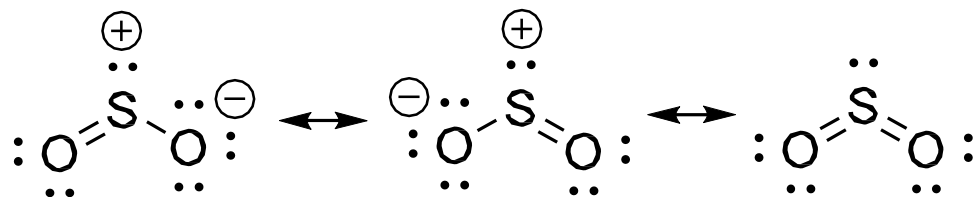
Example: $SN = 3$



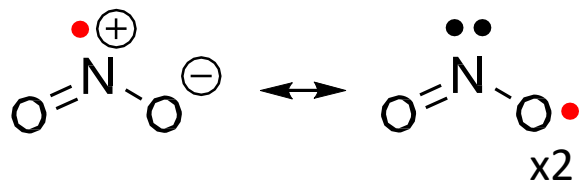
$\text{BF}_3: m = 3, n = 0$ Trigonal planar 平面三角形



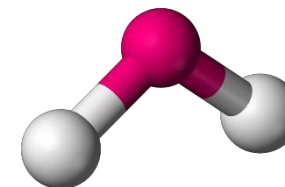
$\text{CH}_3^+: m = 3, n = 0$ Trigonal planar



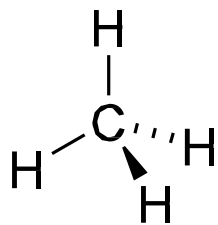
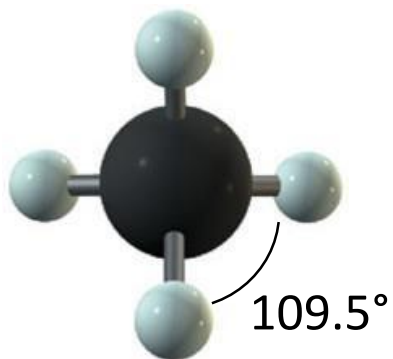
$\text{SO}_2: m = 2, n = 1$ Bent



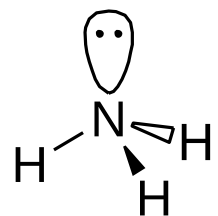
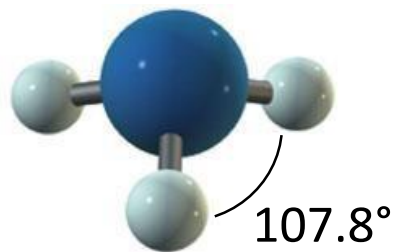
$\text{NO}_2: m = 2, 0.5 < n < 1$ Bent



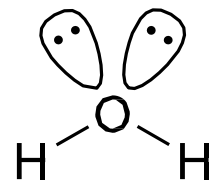
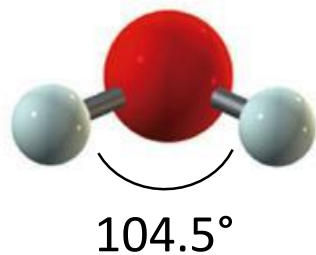
Example: SN = 4



CH₄: $m = 4, n = 0$ Tetrahedral 四面体

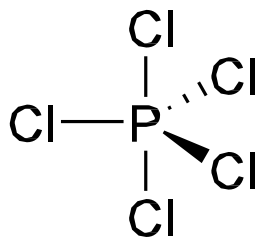


NH₃: $m = 3, n = 1$ Trigonal pyramidal
三角锥

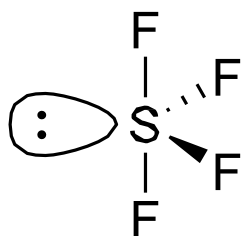
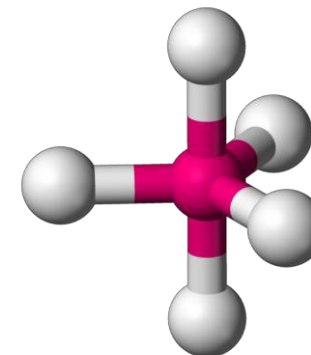


H₂O: $m = 2, n = 2$ Bent

Example: SN = 5

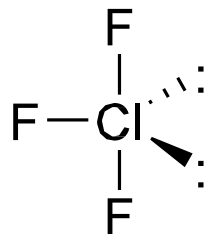
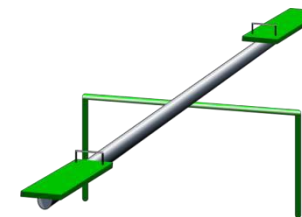


$\text{PCl}_5: m = 5, n = 0$ Trigonal bipyramidal
三角双锥

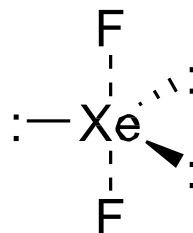
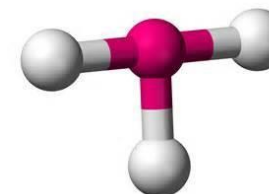


$\text{SF}_4: m = 4, n = 1$ Seesaw

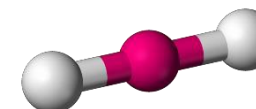
跷跷板



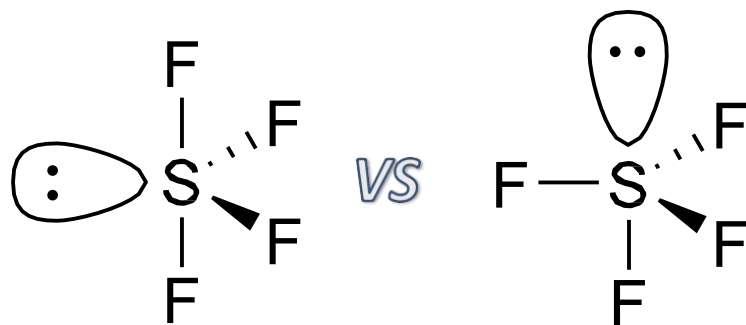
$\text{ClF}_3: m = 3, n = 2$ T-shaped



$\text{XeF}_2: m = 2, n = 3$ Linear

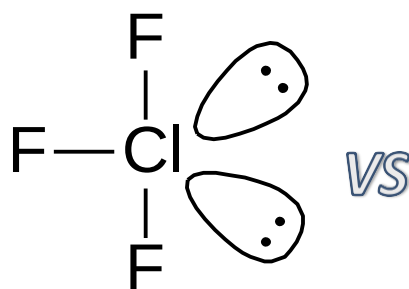
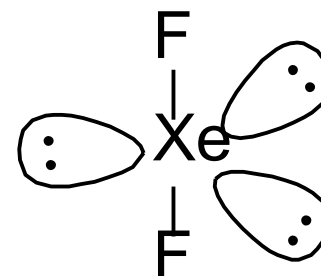


Lone electron pairs

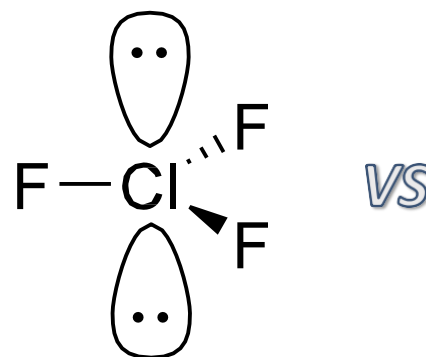


E-X 90° 2 pairs
X-X 90° 4 pairs

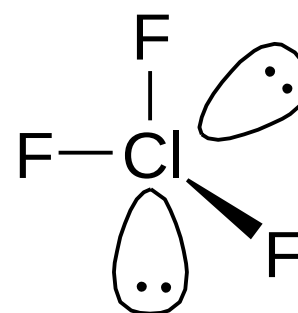
E-X 90° 3 pairs
X-X 90° 3 pairs



E-E 90° 0 pairs
E-X 90° 4 pairs
X-X 90° 2 pairs

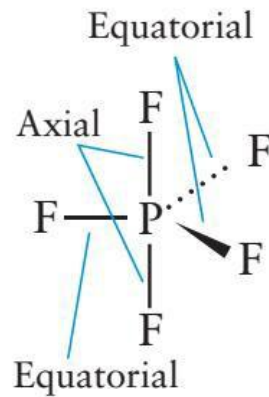


E-E 90° 0 pairs
E-X 90° 6 pairs
X-X 90° 0 pairs



E-E 90° 1 pair
E-X 90° 3 pairs
X-X 90° 2 pairs

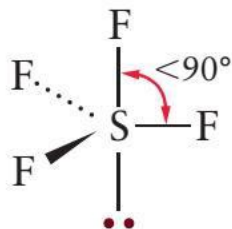
Molecular structures



(a) PF_5



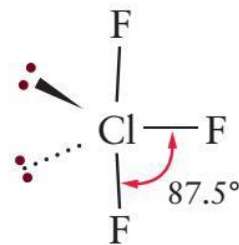
Trigonal bipyramid



(c) SF_4



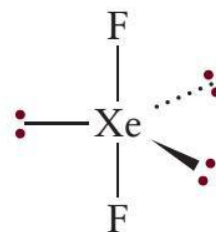
Distorted pyramid
(high-energy, not favored)



(d) ClF_3



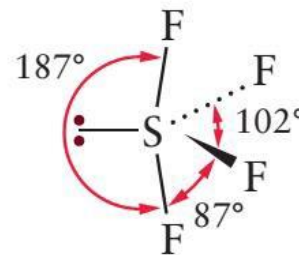
Distorted T



(e) XeF_2



Linear



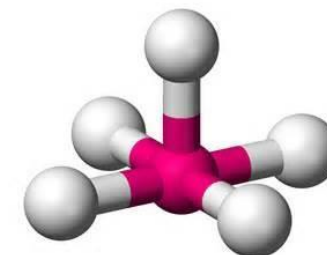
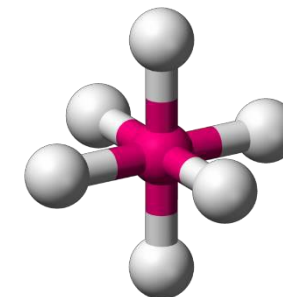
(b) SF_4



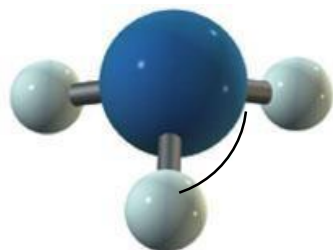
Seesaw
(low-energy, favored)

Summary of molecular geometry

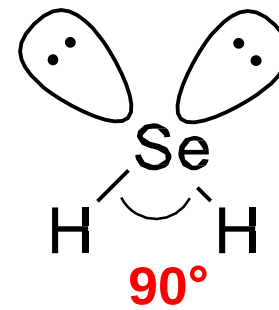
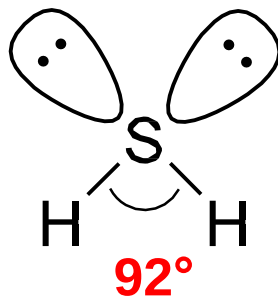
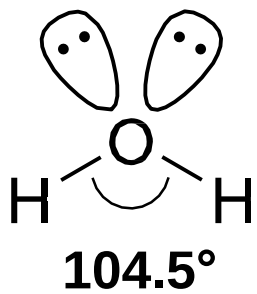
SN of A	Molecular type X is a bonded atom; E is a lone pair	Predicted shape
2	AX_2	Linear (线形)
3	AX_3	Trigonal planar (三角形)
	AX_2E	Bent (弯曲形)
4	AX_4	Tetrahedral (四面体)
	AX_3E	Trigonal pyramidal (三角锥)
	AX_2E_2	Bent
5	AX_5	Trigonal bipyramid (三角双锥)
	AX_4E	Distorted see-saw (跷跷板)
	AX_3E_2	Distorted T (T字)
	AX_2E_3	Linear
6	AX_6	Octahedral (八面体)
	AX_5E	Square pyramidal (四角锥)
	AX_4E_2	Square planar (正方平面)



Lone Pairs Are Fat

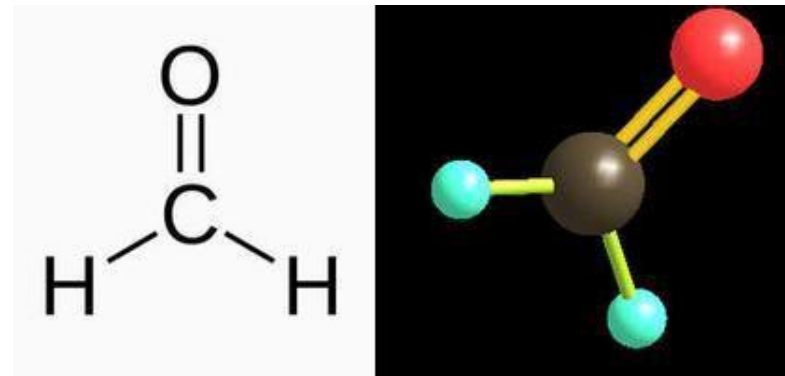


PCl_3	PBr_3	PI_3
100.1°	100.5°	102°



Double bond or triple bond are fat

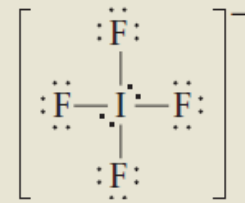
Molecules	Angle of XCX	Angle of XCO
$\text{F}_2\text{C}=\text{O}$	112.5°	123.2°
$\text{Cl}_2\text{C}=\text{O}$	111.3° (size effect)	124.3°
$\text{H}_2\text{C}=\text{O}$	118°	121°
$(\text{NH}_2)_2\text{C}=\text{O}$	118°	121°



Calculate steric numbers for iodine in IF_4^- and for bromine in BrO_4^- . These molecular ions have central I or Br surrounded by the other four atoms.

Solution

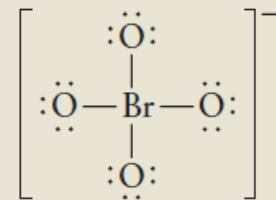
The central I^- atom has eight valence electrons.



Each F atom has seven valence electrons of its own and needs to share one of the electrons from I^- to achieve a noble-gas configuration. Thus, four of the I^- valence electrons take part in covalent bonds, leaving the remaining four to form two lone pairs. The steric number is given by

$$\text{SN} = 4 \text{ (bonded atoms)} + 2 \text{ (lone pairs)} = 6$$

In BrO_4^- , each oxygen atom needs to share two of the electrons from Br^- to achieve a noble-gas configuration.



Because this assignment accounts for all eight of the Br^- valence electrons, there are no lone pairs on the central atom and

$$\text{SN} = 4 \text{ (bonded atoms)} + 0 \text{ (lone pairs)} = 4$$

Predict the geometry of the following molecules and ions: (a) ClO_3^+ , (b) ClO_2^+ , (c) SiH_4 , (d) IF_5 .

Solution

- (a) The central Cl atom has all of its valence electrons (six, because the ion has a net positive charge) involved in bonds to the surrounding three oxygen atoms and has no lone pairs. Its steric number is 3. In the molecular ion, the central Cl should be surrounded by the three O atoms arranged in a trigonal planar structure.
- (b) The central Cl atom in this ion also has a steric number of 3, comprising two bonded atoms and a single lone pair. The predicted molecular geometry is a bent molecule with an angle somewhat less than 120° .
- (c) The central Si atom has a steric number of 4 and no lone pairs. The molecular geometry should consist of the Si atom surrounded by a regular tetrahedron of H atoms.
- (d) Iodine has seven valence electrons, of which five are shared in bonding pairs with F atoms. This leaves two electrons to form a lone pair, so the steric number is 5 (bonded atoms) + 1 (lone pair) = 6. The structure will be based on the octahedron of electron pairs from Figure 3.21, with five F atoms and one lone pair. The lone pair can be placed on any one of the six equivalent sites and will cause the four F atoms to bend away from it toward the fifth F atom, giving the distorted structure shown in Figure 3.24.

Predict whether the molecules NH_3 and SF_6 will have dipole moments.

Solution

The NH_3 molecule has a dipole moment because the three N—H bond dipoles add to give a net dipole pointing upward to the N atom from the base of the pyramid that defines the NH_3 structure. The SF_6 molecule has no dipole moment because each S—F bond dipole is balanced by one of equal magnitude pointing in the opposite direction on the other side of the molecule.

Homework: read “principle of modern chemistry”, chapter 6